

Do Corporate Disclosures Constrain Strategic Analyst Behavior?

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We show that analyst behavior changes in response to a randomly assigned shock that exogenously varies the timeliness and cost of accessing mandatory disclosures in the cross-section of investors: analysts reduce coverage and issue less optimistic, more accurate, less bold, and less informative forecasts. Our evidence indicates that analysts reduce a strategic component of their behavior: the changes are stronger among analysts with more strategic incentives like affiliated or retail-focused analysts. We conclude that mandatory disclosure can substitute for analyst information production, which is constrained by investors' ability to verify forecasts using corporate filings. (*JEL* G18, G29, G38, M41, M48)

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Mandatory disclosure is the cornerstone of U.S. securities market regulation. Major policy changes in disclosure regulation, such as the 2000 Regulation Fair

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Disclosure (Reg FD) or the 2002 Sarbanes-Oxley Act (SOX), aim to improve market quality and protect investors. Yet the optimal level of mandatory disclosure remains hotly debated (Goldstein and Yang 2017), with companies often favoring a lower disclosure burden and investor advocates arguing for greater transparency. How mandatory disclosure shapes a firm's external information environment remains a question of great interest to both scholars of information economics and policy makers (Leuz and Wysocki 2016).

We explore how mandatory disclosure affects analyst behavior and, by implication, investor utility. Sell-side analysts serve an important role as information intermediaries in the stock market, yet a large literature documents biases in analysts' earnings forecasts linked to their strategic incentives.¹ We ask how access to mandatory disclosures, by reducing investors' costs of verifying analyst reports *ex post*, constrains analysts' *ex ante* strategic behavior in the repeated game analysts play with investors.

To identify the causal interplay between access to mandatory disclosures, verification costs, and analyst behavior, we exploit a randomly assigned shock that exogenously reduces the cost of accessing companies' mandatory disclosures for a large subset of investors: the staggered implementation of the SEC's Electronic Data Gathering, Analysis, and Retrieval (EDGAR) system. Before EDGAR, investors could access firms' mandatory filings only at high cost, either by subscribing to commercial data feeds or by physically visiting one of the SEC's reference rooms in Chicago, New York, or Washington, DC (Rider 2001). As a result, only the largest institutional investors had timely access to corporate filings. Beginning in April 1993, the SEC required U.S. firms to file their mandatory disclosures (such as 10-Ks, 10-Qs, or 8-Ks) electronically through the EDGAR system, giving all investors free and equal access. As we show, making a firm's SEC filings available through EDGAR reduced asymmetries of information access among market participants, as measured by standard proxies for trading liquidity, without changing the nature of the information firms disclose in their filings.

Helpful for identification purposes, the SEC *randomly* assigned firms to 1 of 10 phase-in waves, thereby staggering inclusion in EDGAR over a 3-year period between 1993 and 1996.² We can thus compare firms that were randomly included in EDGAR in quarter *t* to observably similar control firms not yet included in EDGAR. Random assignment and staggered implementation reduce endogeneity concerns. Critically, an omitted variable would need to coincide in time with the phase-in dates to materially confound our findings. Equally helpfully, the SEC changed key features of the rollout in

¹ Prior work has shown that analyst forecasts affect stock prices (Womack 1996; Gleason and Lee 2003; Jegadeesh et al. 2004; Ljungqvist, Malloy, and Marston 2009; Kelly and Ljungqvist 2008). For recent surveys of the literature on analysts' strategic behavior, see Beyer et al. (2010), Bradshaw (2011), and Kothari, So, and Verdi (2016).

² Appendix A lists the 10 phase-in dates. In private correspondence, Scott Bauguess, then-Acting-Chief-Economist of the SEC, informed us that the wave assignments were randomized conditional on firm size.

ways that imply that a firm's inclusion in EDGAR can be viewed as a surprise, reducing concerns that firms, analysts, or investors altered their behavior in anticipation.

Using a stacked difference-in-differences approach, we show that analysts change their behavior when a firm joins EDGAR and its filings thereby become freely and universally available to all investors. First, we find a reduction in analyst coverage, suggesting that mandatory disclosure and information production by analysts are substitutes to some extent.³ Second, we find large improvements in the two most widely studied measures of analyst behavior, inaccuracy and optimism (O'Brien 1988): after a firm joins EDGAR, analysts make forecasts that are significantly more accurate and less optimistic about the firm's prospects than before, both as a group (i.e., at the stock level) and individually (i.e., at the analyst/stock level). Third, we find that forecasts have less price impact post-EDGAR, suggesting that broader access to SEC filings improves firms' information environments and so reduces the informativeness of forecasts. This pattern reinforces our conclusion that mandatory disclosure and information production by analysts are substitutes to some extent. Finally, dispersion in analysts' forecasts and their willingness to deviate from other analysts both decline post-EDGAR.

Why do analysts change their behavior when a firm joins EDGAR? Much of the literature views optimistic forecasts as strategic, arising from analysts' career concerns, conflicts of interest from the investment banking or trading divisions, or a desire to cater to the management of firms analysts follow.⁴ An influential strand of the theory literature models the strategic interaction between analysts and investors as a cheap-talk game in which investors obtain information from self-interested analysts who may or may not bias their reports. A key insight of this literature is that when interests are not aligned, analysts will not truthfully report their signals (Crawford and Sobel 1982) even though they care about their reputations (Benabou and Laroque 1992) and investors can impose reputation costs by punishing misreporting (Bolton, Freixas, and Shapiro 2012).⁵ Fischer and Stocken (2010) model how a change in the precision of a nonstrategic public signal (such as a firm's mandatory disclosures) affects analysts' incentives to gather and report information. When an analyst's credibility is in doubt, an increase in the precision of the public signal induces the analyst to report more accurate

³ This result is reminiscent of the fall in analyst coverage following Reg FD (Irani and Karamanou 2003), which has been interpreted as a crowding-out effect of increased mandatory disclosure.

⁴ For work on career concerns leading to forecast optimism, see Stickel (1992), Mikhail, Walther, and Willis (1999), and Hong, Kubik, and Solomon (2000). For work on conflicts of interest arising from investment banking or trading commissions, see McNichols and O'Brien (1997), Michaely and Womack (1999), Ljungqvist, Marston, and Wilhelm (2006), and Groysberg, Healy, and Maber (2011). For work on incentives related to catering to management, see Francis and Philbrick (1993), Das, Levine, and Sivaramakrishnan (1998), Chen and Matsumoto (2006), Mayew (2008), and Hilary and Hsu (2013).

⁵ In the present context, punishment may take the form of investors moving their brokerage accounts elsewhere when they learn that an analyst has biased her forecasts.

information in equilibrium, in an effort to get investors to trade more in response to her report. Bolton, Freixas, and Shapiro (2012) reach a similar conclusion with regards to bias, noting that greater signal precision increases the probability of an analyst getting caught ex post and being punished for misreporting.

EDGAR can be viewed as increasing the precision of the public signal (in the sense that investors gain free and universal access to the full text of firms' standardized financial disclosures rather than having to rely on unregulated summaries in the form of earnings releases) and thus in the risk of detection and punishment. If so, we expect that once a firm joins EDGAR, analysts moderate their strategic behavior by increasing forecast accuracy (Fischer and Stocken 2010) and reducing forecast bias (Bolton, Freixas, and Shapiro 2012). Our findings are consistent with both predictions.

In the pre-EDGAR era, access to corporate filings was prohibitively costly for all but the largest investors. Retail investors thus experienced a relatively larger reduction in the cost of accessing corporate filings thanks to EDGAR. We predict and find that analysts who cater to retail clients moderate their behavior by more when a stock joins EDGAR than analysts who serve institutional clients. Moreover, we find more pronounced changes in behavior among analysts with stronger incentives to strategically skew their forecasts, such as analysts with ties to management.

We find no support for two alternative explanations. First, we take seriously the possibility that universal access to corporate filings could reduce analysts' own information production costs, even though this effect is likely small as most brokerage firms already had access to SEC filings (Christensen, Heninger, and Stice 2013).⁶ We find no evidence to support this broker channel. In particular, it is not the case that forecast accuracy improves by more at brokers that are more likely to have experienced a reduction in information production costs. Second, we find no evidence that firm fundamentals or transparency change post-EDGAR, which reduces concerns that the observed changes in analyst behavior simply reflect changes in firms' prospects or disclosure policies.

We conclude that greater access to mandatory disclosures improves the ability of retail and small institutional investors to verify analysts' reports ex post with the help of the standardized financial information available in SEC filings. Improved ex post verification increases the ex ante risk of ex post detection for analysts who strategically misreport their signals, constraining analyst behavior ex ante.

⁶ Interviews with veteran analysts and a head of equity research confirm that in the pre-EDGAR era, brokers typically had access to all corporate filings through one or more of the many data feeds available via the Bloomberg or Reuters terminal. These feeds included the Q-Data Company's "SEC File," Disclosure Information Group's "Compact Disclosure," Dialog's "SEC ONLINE," McGraw-Hill's "COMPSTAT Corporate Text," and Mead Data Central's Lexis-Nexis service. Analysts at brokers without comprehensive data feeds could get filings faxed to them by a company's investor relations department.

At the same time as analysts moderate their forecast biases post-EDGAR, we show that they begin to issue significantly more bullish stock recommendations. A plausible interpretation of this divergence between forecasts and recommendations is that analysts shift their management-pleasing behavior from friendly forecasts, which EDGAR makes easier to detect, to friendly recommendations, which remain fundamentally subjective and difficult to verify *ex post* (Lin and McNichols 1998). Consistent with this interpretation, we find that investors become more skeptical of recommendation upgrades post-EDGAR.

Our study makes contributions to two literature. First, we contribute to the mandatory disclosure literature. Sellside analysts are viewed as key information intermediaries whose self-serving strategic behavior affects the quality of firms' external information environments. Our evidence suggests that free, timely, and equal access to mandatory disclosures can enable investors to police analysts' strategic behavior in ways that improve firms' information environments, resulting in improved trading liquidity and thereby, implicitly, reductions in their cost of capital. Our conclusion that equal *access* to mandatory disclosures matters complements prior work focusing on the *content* of mandatory disclosures, often through the lens of regulatory changes, such as Reg FD or SOX (Chhaochharia and Grinstein 2007; Duarte et al. 2008; Koch, Lefanowicz, and Robinson 2013; Coates and Srinivasan 2014).

Our evidence contributes to the debate on the costs and benefits of mandatory disclosure. On the plus side, we identify a previously unknown benefit of disclosure: it can help investors police analyst behavior. Improved analyst behavior can reduce investor disagreement (Xiong 2013), which in turn can reduce overpricing and crash risk (Chang et al. 2022). On the minus side, we show that universal access to corporate filings leads to reductions in analyst coverage and the informativeness of forecasts. This reinforces concerns that disclosure regulations, motivated by a desire to level the informational playing field, can have unintended consequences (Gintschel and Markov 2004; Gomes, Gorton, and Madureira 2007; Koch, Lefanowicz, and Robinson 2013). Crowding out information production can adversely affect investors both directly, as information asymmetries increase in markets, and indirectly, as fewer analysts monitor firms and managers and corporate governance weakens (Jensen and Meckling 1976).

We hope that our policy experiment can expand the methodological range of natural experiments with which the costs and benefits of disclosure can be investigated. Reviewing the disclosure regulation literature, Leuz and Wysocki (2016) propose that "identification and causal inferences are of first-order importance for policy and regulatory debates." As we argue, the staggered implementation of EDGAR, along with random assignment, significantly reduces endogeneity concerns, not least compared to the two workhorse shocks considered in the literature: the adoption of IFRS accounting standards in

various countries outside the United States and the 2002 Sarbanes-Oxley Act in the United States.⁷

The second literature we contribute to is the literature on analyst behavior. Our central conclusion that giving investors more equal access to corporate filings constrains analysts' strategic forecasting behavior, but not their strategic recommendations behavior, highlights the importance of ex post verification of analyst reports: while the accuracy of earnings forecasts can easily be verified ex post once investors have access to corporate filings, buy and sell recommendations cannot, given that investment horizons are left vague. Ex post verification complements the other constraint on strategic analyst behavior studied in the literature: reputation.⁸ An increased threat of ex post verification increases the expected reputational cost of self-serving forecasts and so moderates analyst behavior.

Our paper is part of a recent body of work exploiting the staggered way in which EDGAR was implemented. We differ from this body of work in that we focus on the game played between analysts and investors.⁹ We disentangle the effects of EDGAR on analysts' strategic behavior and on their information production and establish that the ability of investors to verify analyst reports ex post affects analysts' strategic behavior ex ante.

1. Empirical Strategy and Data

1.1 Institutional background

Testing how access to SEC filings affects strategic analyst behavior requires a shock to disclosure access that is randomly assigned to some firms, whereas others are unaffected and so can serve to establish a counterfactual. Our identification strategy relies on the introduction of the SEC's EDGAR system. To understand how EDGAR made the informational playing fields between investors and analysts and among investors more level, consider how investors accessed corporate filings in the pre-EDGAR era.

Prior to EDGAR, firms subject to SEC registration were required to mail their mandatory filings in hardcopy to the SEC. To access filings,

⁷ Leuz and Wysocki (2016) conclude that "few studies [...] are able to attribute the documented effects to IFRS adoption" as "IFRS were often adopted amidst a series of other (unrelated) institutional reforms." SOX, which affected all listed U.S. firms at more or less the same time, is similarly "susceptible to confounding effects by concurrent events."

⁸ See Hong, Kubik, and Solomon (2000), Krigman, Shaw, and Womack (2001), Cowen, Groyberg, and Healy (2006), Ljungqvist, Marston, and Wilhelm (2006), Ljungqvist et al. (2007), Clarke et al. (2007), and Kolasinski and Kothari (2008).

⁹ The first to exploit the staggered implementation of EDGAR are Gao and Huang (2020), who view EDGAR as a technology shock that facilitates better information dissemination. Their main finding is that individual investors' trades become more informative of future returns after EDGAR inclusion, a result that sits well with our claim that better access to information in the form of mandatory disclosures improves investors' ability to police analyst behavior. Chang et al. (2022) explore EDGAR's asset pricing consequences in the context of disagreement models, finding that EDGAR inclusion leads to stock price corrections around information events and reduces stock price crash risk as a result of a reduction in disagreement among investors.

investors could either visit an SEC reference room (in Chicago, New York, or Washington, DC) or obtain electronic copies via a commercial data vendor.¹⁰ Commercial access was, apparently, quite costly. According to a 1992 petition to the SEC signed by academics, librarians, and journalists, Mead Data Central charged “a fee of \$125 per month, plus a connect charge of \$39 an hour, plus a charge of 2.5 cents per line of data plus search charges which range from \$6 to \$51 per search”; Dialog, a competitor to Mead, charged “\$84 per hour plus \$1 per page.”¹¹ To illustrate, obtaining Ford’s 1994 10-K from Dialog would have cost \$145 in page charges alone.

Given these access options, there were three categories of investors: those who chose not to have access to mandatory filings, those who accessed them physically (likely with some delay), and those who paid for timely online access. We suspect that most individuals and quite a few institutional investors fell into the no-access category, with only those located near an SEC reference room accessing filings physically, and only larger institutions paying for online access. As a result, we conjecture that there were widespread and systematic informational asymmetries across different investor groups pre-EDGAR. Retail investors in particular were at an informational disadvantage, not only relative to institutional investors but also relative to information intermediaries, such as sell-side analysts.

On February 23, 1993, the SEC announced a plan to require all SEC-registered firms to submit their filings electronically. SEC Release No. 33-6977 included a draft phase-in schedule, with registered firms joining EDGAR in 10 waves over the 3 years starting April 26, 1993, and ending May 6, 1996. Firms in waves 5 through 10 did not know their EDGAR join dates until a few months before joining.¹²

The SEC originally planned to allow public access to EDGAR only via dedicated terminals located in its three reference rooms. As a result, electronic *filing per se* would not have affected investors’ costs of accessing mandatory disclosures. The actual shock to information *access* that we exploit is due to the National Science Foundation’s decision in October 1993 to acquire Mead Data Central’s historic EDGAR filings and to fund a project to make EDGAR filings available for free online, hosted by NYU. Online access to EDGAR went live on January 17, 1994, when the historic and current filings of firms in the SEC’s first four phase-in waves (as well as those of previous voluntary filers) became

¹⁰ Investors who owned shares on the record date could wait to receive a copy of the annual report in the mail. However, 10-Ks provide information not provided in annual reports (Asthana and Balsam 2001).

¹¹ Quoted from <http://www.bio.net/bionet/mmm/ag-forst/1992-January/000187.html>.

¹² The draft phase-in schedule included a 6-month review, to begin after wave 4 on December 6, 1993. The review took longer than planned, leading to the suspension of waves 5 (previously scheduled for Aug. 1994) and 6 (previously scheduled for Nov. 1994). On December 19, 1994, the SEC’s final rule revised the dates for waves 5 and 6 to January 1995 and March 1995, confirmed the date for wave 7, and finalized the dates for waves 8 through 10 (SEC Release No. 33-7122). We use the final (i.e., actual) phase-in dates as per the December 1994 release. In this regard, we follow Goldstein, Yang, and Zuo (2020) and Chang et al. (2022) but depart from some prior work, such as Gao and Huang (2020), using the EDGAR shock.

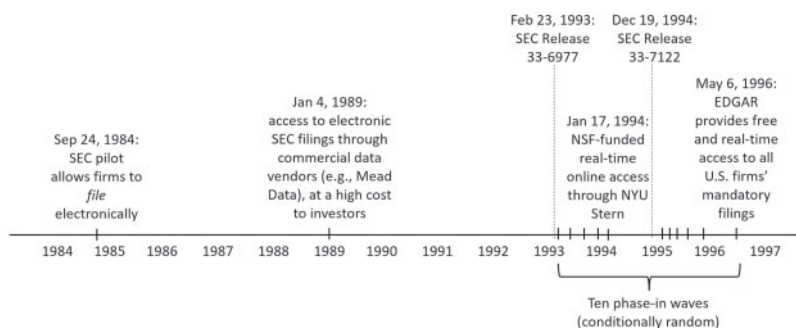


Figure 1
Timeline of EDGAR implementation

The figure shows the major milestones in the SEC's implementation of EDGAR. SEC Release 33-6977 is the SEC's announcement of its plan to require all registered firms to submit their filings electronically, ultimately in 10 waves. The release contains the phase-in dates for four "significant test groups," to be followed by a 6-month evaluation period in the first half of 1994 leading to a final rule concerning the phase-in dates for the remaining firms. SEC Release 33-7122 contains final rules on EDGAR implementation, including the dates of the remaining six waves. The National Science Foundation announced on October 22, 1993, funding for a project to make all EDGAR filings available for free online, hosted by New York University's Stern School of Business. The SEC took over online access in October 1995.

available via the NYU online-access system.¹³ In the six remaining waves, firms became electronic EDGAR filers and had their historic and current filings become publicly available online at the same time. Figure 1 illustrates the timeline of events. (Unless otherwise noted, we refer to the introduction of online access to corporate filings via first NYU and eventually the SEC's EDGAR website as "EDGAR inclusion.")

1.2 Identification strategy

Our identification strategy exploits three features of the way the SEC implemented EDGAR. First, the SEC assigned registered firms to the 10 phase-in waves *randomly*, conditional only on size. Second, while all registered firms joined EDGAR eventually, the staggered rollout of EDGAR provides a set of control firms with which to establish a counterfactual that is plausibly free of the confounding effects of unobserved contemporaneous factors that might have affected analyst behavior. Confounds would have to not only coincide in time with the EDGAR phase-in schedule (and the NSF's online access timetable) but also affect treated firms (but not controls) at around the same time as their filings became available online, which, while not impossible, strikes us as unlikely. Third, the fact that firms in waves 1–4 did not know that their filings were ever going to be put online, coupled with the fact that firms in waves 5–10 were only given short notice of their phase-in dates, greatly reduces the risk of confounds that result from firms, analysts, or investors changing their behavior ahead of treatment.

¹³ The SEC took over the task of hosting online access to EDGAR from NYU in October 1995.

Our identification strategy uses a difference-in-differences (DD) design, comparing a set of treated firms whose mandatory disclosures become freely available to all investors at the same time to a set of matched control firms with similar characteristics whose disclosures randomly remain expensive for investors to access. Specifically, we estimate the following stacked DD regression (Cengiz et al. 2019):

$$outcome_{iwt} = \sum_k \beta_k \mathbb{I}(t - T_{iw} = k) + \gamma \mathbf{X}_{iwt-1} + c_{iw} + c_q + c_{if} + \varepsilon_{iwt} \quad (1)$$

where $outcome_{iwt}$ in quarter t is measured for firm i joining EDGAR in phase-in wave w at time T_{iw} ; $\mathbb{I}(t - T_{iw} = k)$ is a treatment indicator for being k quarters from the EDGAR joining quarter; $\mathbf{X}_{iwt-1}$ is a vector of control variables; and c_{iw} , c_q , and c_{if} are wave-specific firm, calendar-time, and fiscal-quarter fixed effects. Standard errors are clustered at the firm level, given that we exploit a firm-level shock and the time dimension of our panel is substantially smaller than the firm dimension (Petersen 2009; Section 3).^{14,15} Random assignment and staggering go a long way toward ensuring the internal validity of the EDGAR experiment. We discuss the plausibility of the identifying assumptions behind our DD design in detail in Section 2.3.

1.3 Sample and data

1.3.1 Treated and control firms. To ensure uniform reporting and disclosure standards, we restrict the sample of treated and control firms to firms traded on the NYSE, NASDAQ, or AMEX and exclude firms with CRSP share codes greater than 11 (foreign issuers, real estate trusts, master limited partnerships, and the like).

With one important exception, firms are treated from the fiscal quarter in which they are included in EDGAR. The exception concerns firms in phase-in waves 1 through 4, whose electronic EDGAR filings did not become publicly available *online* until January 17, 1994, and so are considered treated only from that date onward.¹⁶ Eventually, all SEC-registered firms are treated, as every issuer is obliged to file through EDGAR starting on May 6, 1996. To avoid biases that can arise in staggered DD designs with time-varying treatments

¹⁴ Our results are robust to double clustering by firm and fiscal quarter instead, consistent with Petersen's (2009, p. 460) conclusion that "[w]hen there are only a few clusters in one dimension, clustering by the more frequent cluster yields results that are almost identical to clustering by both firm and time." In our setting, robustness is unsurprising: given conditionally random assignment, firms in each phase-in wave have only one thing in common, namely, size. Since our regressions control for size, there are unlikely to be unobserved characteristics that could induce correlated responses within a quarter.

¹⁵ We also consider a variation of Equation (1) estimated at the analyst-stock level, in which $outcome_{ikt}$ is measured for the firm i -analyst k pair in fiscal quarter t and the firm fixed effects are replaced with firm-by-analyst fixed effects. In this variation, we double-cluster the standard errors at the firm and the analyst-quarter level.

¹⁶ This is another respect in which we follow Chang et al. (2022) but depart from much prior work using EDGAR as a shock, such as Gao and Huang (2020). As we will show in Section 3.2, analysts respond in ways that support our claim that it is online access, not electronic filing, that matters.

and treatment effect heterogeneity (Baker, Larcker, and Wang 2022), we select “clean” control firms from the set of future-treated firms, ensuring that control firms are not themselves subject to treatment while serving as controls. Naturally, the last EDGAR wave lacks clean controls and (because of bunching toward the end of the SEC’s phase-in schedule) so do waves 8 and 9. This leaves us with four staggered treatment dates: January 17, 1994, January 30, 1995, March 6, 1995, and May 1, 1995.

We select control firms using a nearest-neighbor propensity-score method. Given that our DD models exploit a staggered event, we create cohort-specific data sets for each wave (Baker, Larcker, and Wang 2022; Cengiz et al. 2019). Specifically, we estimate one propensity-score model for each of the staggered treatment dates for which controls can be guaranteed to be clean. We match treated and controls on three dimensions: equity market cap (in levels and logs), to hold constant the SEC’s size criterion when assigning firms to phase-in waves; fiscal quarter, to hold constant well-known seasonalities in analyst forecasts over the course of the fiscal year;¹⁷ and the logarithm of the number of analysts covering a stock, to hold constant competitive effects constraining analyst behavior (Hong and Kacperczyk 2010).¹⁸ To ensure covariate balance, only matches in the common support are considered valid, using a 0.05 caliper. This limits our sample to a total of 2,016 treated and 2,016 control firms.

As Appendix A shows, the average sample firm has an equity market capitalization of \$171.7 million in the quarter before treatment, an amount that is considerably less than the \$879.6 million market cap of the average listed U.S. firm in the quarter before its phase-in wave. Appendix A shows why. The SEC skewed assignment in the first two waves toward large firms. Because the first two waves occurred only 3 months apart, few large untreated firms are left in the common support: only 68 of the 510 firms in the first two waves have valid clean controls. To the extent that smaller firms provide analysts greater scope to engage in strategic behavior, our empirical estimates may accordingly overstate the effects of universal access to SEC filings on analyst behavior for the average U.S.-listed firm.

We follow each treated firm and its matched control for nine fiscal quarters centered on the quarter the treated firm’s filings went online (its EDGAR inclusion quarter for short), except in wave 7, for whose fourth post-EDGAR quarter there are no clean controls. Coupled with the fact that we use clean controls, our research design is equivalent to Cengiz et al.’s (2019) stacked-regression estimator, which Baker, Larcker, and Wang (2022) show, using simulations, to be unbiased.

¹⁷ Earnings forecasts tend to become more accurate the later in a firm’s fiscal year they are made (Richardson, Teoh, and Wysocki 2004). See Chang et al. (2022) for further discussion.

¹⁸ Table IA.1 in the Internet Appendix reports the propensity-score estimates. All our results are robust to using any reasonable matching model that captures the size criterion of the SEC’s conditionally random assignment to treatment.

1.3.2 Measures of analyst behavior. Following the extensive literature on analyst behavior, we focus on analyst optimism (or forecast bias), inaccuracy (or forecast errors), the informativeness (or price impact) of forecast revisions and recommendation changes, dispersion in forecasts, and forecast boldness (or deviations from consensus). Analysts make short- and long-term forecasts. Accordingly, we measure the optimism, inaccuracy, dispersion, and boldness of current-quarter forecasts for both next-fiscal-quarter earnings and fiscal-year earnings. Appendix B offers variable definitions and the details of their construction.

Table 1 reports summary statistics. Treated and control firms have near-identical optimism, inaccuracy, informativeness, dispersion, and boldness in the quarter before treatment, both in levels and in changes. Simple *t*-tests confirm that with one exception, the difference in pretreatment changes between treated and controls is insignificant. The exception is optimism in long-term forecasts, which increases by significantly more for control firms than for treated firms in the quarter before treatment.¹⁹

1.3.3 Control variables. Given random assignment, treated and control firms differ only randomly in their characteristics. While this obviates the need for the kinds of control variables sometimes included in empirical work in this area, we still have to deal with two issues. The first is that the SEC's assignment to treatment is *conditionally* random. Our research design takes this issue into account by matching on market cap. As Table 1 shows, treated and controls are matched quite precisely on market cap. We additionally include the logarithm of the market cap as a control variable in Equation (1), following the advice of Imbens (2004). The second issue is the aforementioned seasonality in analyst forecasts. To hold seasonality constant, our research design matches on fiscal year-end when selecting control firms. We additionally include fixed effects for fiscal quarter in our DD regressions.

Finally, we include the usual time and firm fixed effects in our specifications, to ensure consistent estimation of treatment effects in a DD context. Since time is measured in quarters in our setting, we include calendar-quarter fixed effects. These time effects remove the effects of any common shocks that affect all firms in a given quarter, such as marketwide events or macroeconomic news.

1.4 EDGAR filings versus earnings releases

Research on what is called the value relevance of corporate filings shows that 10-K and 10-Q filings had little effect on share prices or trading volume in the 1970s and 1980s (Stice 1991; Easton and Zmijewski 1993); instead, investors tended to respond to information contained in the press releases accompanying

¹⁹ In our subsequent DD regressions, we cannot reject the null hypothesis of no diverging pre-trends even in long-term optimism, supporting the parallel trends assumption required for identification.

Table 1
Summary statistics

	Pretreatment levels						Pretreatment changes (from $t-2$ to $t-1$)						Treated – controls	
	Treated firms			Control firms			Treated firms			Control firms			Difference	t -stat
	# obs.	Mean	SD	# obs.	Mean	SD	# obs.	Mean	SD	# obs.	Mean	SD		
Matching variables:														
Market capitalization (\$m)	2,016	171.7	394.8	2,016	193.3	407.4	2,008	4.922	83.878	1,996	3.224	67.450	1.698	0.706
# of analysts	2,016	2.110	3.216	2,016	2.058	2.953	2,016	-0.017	1.137	2,016	-0.005	1.178	-0.012	-0.326
Analyst behavior:														
Optimism (short-term)	544	0.008	0.026	589	0.006	0.022	393	-0.002	0.037	421	0.001	0.030	-0.003	-1.287
Optimism (long-term)	1,080	0.031	0.090	1,058	0.023	0.079	956	-0.005	0.053	925	0.000	0.055	-0.005	-2.130
Inaccuracy (short-term)	544	0.012	0.024	589	0.011	0.020	393	-0.004	0.032	421	-0.002	0.026	-0.002	-0.997
Inaccuracy (long-term)	1,080	0.042	0.091	1,058	0.034	0.080	956	-0.006	0.052	925	-0.001	0.053	-0.005	-1.861
Informativeness	2,016	0.036	0.058	2,016	0.036	0.057	2,016	-0.002	0.047	2,016	-0.001	0.045	-0.001	-0.887
Revision response coeff.	2,016	0.228	4.706	2,016	0.085	5.138	2,016	0.064	5.958	2,016	0.142	6.735	-0.078	-0.391
Price impact rec. changes	2,016	0.001	0.020	2,016	0.002	0.023	2,016	0.000	0.029	2,016	0.001	0.031	-0.001	-1.188
Dispersion (short-term)	504	0.003	0.004	530	0.002	0.003	438	0.000	0.003	477	0.000	0.002	0.000	-0.259
Dispersion (long-term)	937	0.009	0.014	1,013	0.007	0.013	902	0.000	0.009	984	0.000	0.010	0.000	-0.272
Boldness (short-term)	181	0.004	0.009	190	0.004	0.009	102	0.002	0.008	109	0.002	0.010	0.000	-0.135
Boldness (long-term)	534	0.008	0.023	543	0.005	0.009	385	-0.001	0.019	405	-0.001	0.016	-0.001	-0.412
Market reaction:														
Abnormal volume	2,013	1.277	1.063	2,015	1.242	1.024	2,011	-0.005	1.605	2,015	-0.065	1.681	0.060	1.166
Abnormal volume (retail)	867	0.005	0.072	737	0.007	0.079	751	0.008	0.116	618	0.005	0.136	0.003	0.488
AIM	1,847	1.067	1.291	1,871	1.084	1.252	1,795	-0.054	0.472	1,827	-0.026	0.524	-0.027	-1.645
Effective tick	1,847	0.027	0.029	1,868	0.027	0.028	1,794	0.000	0.017	1,823	0.000	0.018	-0.001	-1.202
Fraction zero-return	2,016	0.275	0.123	2,016	0.254	0.121	2,016	-0.008	0.095	2,016	-0.007	0.091	-0.002	-0.617
Idiosyncratic volatility	2,015	0.038	0.026	2,016	0.042	0.028	2,015	0.001	0.013	2,015	0.001	0.015	0.000	0.629

The table reports summary statistics for the variables used in our analysis, separately for treated and control firms measured in levels and changes in the quarter before treatment. Treated firms are those included in EDGAR; control firms are nearest-neighbor propensity-score matched on equity market capitalization (in levels and logs), analyst coverage (in logs and lags), and fiscal quarter using a 0.05 caliper. All variables are measured at the firm/fiscal-quarter level. All other variables are measured at the firm/fiscal-quarter level. For variable definitions and details of their construction, see Appendix B. The final two columns test whether the difference in pretreatment changes between treated and controls is statistically significant.

firms' earnings announcements. These findings raise the question whether access to corporate filings via EDGAR materially improves investors' information sets. Prior research has suggested that it does. Using event-study designs, Qi, Wu, and Haw (2000), Asthana and Balsam (2001), Griffin (2003), Asthana, Balsam, and Sankaraguruswamy (2004), Li and Ramesh (2009), and Christensen, Heninger, and Stice (2013) report significant market reactions on corporate filing days in the EDGAR era, confirming that broader access to corporate filings through EDGAR makes filings value relevant.²⁰

The value relevance of corporate filings, once all investors can freely access them, is not surprising. As Griffin (2003) notes, "Form 10-K and Form 10-Q filings are unquestionably the most comprehensive and detailed single source of financial information available to stock investors. They often contain highly significant information about company performance and financial position not provided by other means such as earnings announcements." Moreover, corporate filings contain more of the kind of information investors are interested in. McClure and Nikolaev (2022) conclude that investors prefer information about operating profitability rather than the types of "bottom-line earnings" figures highlighted in earnings releases²¹ and that investors are especially interested in disaggregated information about accruals, which is only available in SEC filings.

2. Disclosure Access and Analyst Behavior

2.1 Validating the shock

To establish that EDGAR inclusion is a sufficiently large shock to a firm's information environment such that it has the potential to materially affect analyst behavior, we begin by estimating changes in a standard measure of investor attention, abnormal trading volume (Barber and Odean 2008). Table 2 shows that trading volume increases significantly in the fiscal quarter a firm is included in EDGAR, relative to matched controls not yet included in EDGAR ($p=.016$). The point estimate shown in column 1 suggests that trading volume increases by an economically meaningful 5.6% from the sample mean in the pretreatment quarter. Retail investors should be particularly responsive to easier access to corporate disclosure, as they faced the highest access costs to

²⁰ In the Internet Appendix, we replicate the core finding of these six studies by estimating the price impact of corporate filings in the 3-day window around the 10-K or 10-Q filing dates of EDGAR joiners in our sample. As Figure IA.1 shows, corporate filings have a significant price impact in our sample.

²¹ Managers have wide discretion over how earnings are presented in press releases. Prior research has documented that earnings announcements tend to highlight pro forma (or "Street") earnings, which exclude a variety of expenses under GAAP. This practice only began to change after the SEC issued a warning against non-GAAP measures in 2001 (Bradshaw and Sloan 2002; Bowen, Davis, and Matsumoto 2005), culminating in January 2002 in Regulation G, which requires "public companies that disclose or release such non-GAAP financial measures to include, in that disclosure or release, a presentation of the most directly comparable GAAP financial measure and a reconciliation of the disclosed non-GAAP financial measure to the most directly comparable GAAP financial measure."

Table 2
Validating the shock

	Abnormal volume		Liquidity			Volatility
	All trading (1)	Retail trading (2)	AIM (3)	Effective tick (4)	Fraction zero-return days (5)	Idio-syncratic volatility (6)
Quarter of EDGAR inclusion	0.072** <i>0.030</i>	0.008** <i>0.004</i>	−0.016 <i>0.014</i>	−0.060 <i>0.047</i>	−0.546*** <i>0.219</i>	0.012 <i>0.039</i>
Next four quarters	−0.010 <i>0.028</i>	0.000 <i>0.003</i>	−0.052*** <i>0.018</i>	−0.080* <i>0.048</i>	0.047 <i>0.227</i>	−0.101** <i>0.048</i>
Controls?	Yes	Yes	Yes	Yes	Yes	Yes
Calendar quarter FE?	Yes	Yes	Yes	Yes	Yes	Yes
Fiscal quarter FE?	Yes	Yes	Yes	Yes	Yes	Yes
Firm FE?	Yes	Yes	Yes	Yes	Yes	Yes
R-squared (%)	12.8	10.6	88.8	76.9	69.8	81.4
No. of firms	4,031	2,440	3,915	3,912	4,032	4,031
No. of firm-quarters	33,895	13,879	31,383	31,340	33,924	33,905

The table reports difference-in-differences estimates of the effects of inclusion in EDGAR on abnormal trading volume, trading liquidity, and volatility. Treated firms are those included in EDGAR; control firms are nearest-neighbor propensity-score matched on equity market capitalization (in levels and logs), number of analysts (in logs and lags), and fiscal quarter using a 0.05 caliper. We include data from a nine-fiscal quarter window centered on the fiscal quarter in which a treated firm's EDGAR inclusion takes place. All specifications are estimated using OLS and include controls (the one-quarter lag of log market cap) and fixed effects for calendar-quarter, fiscal-quarter, and firm. For variable definitions and details of their construction, see Appendix B. The coefficients in columns 4, 5, and 6 are multiplied by 100 for ease of exposition. Heteroscedasticity-consistent standard errors clustered at the firm level are shown in italics underneath the coefficient estimates. * $p < .1$; ** $p < .05$; *** $p < .01$.

begin with. Column 2 shows that retail trading volume (measured using Barber and Odean's discount-brokerage data) increases in the fiscal quarter a firm is included in EDGAR ($p = .050$). We interpret these increases in trading activity as consistent with investors paying more attention to a firm when its mandatory disclosures are more easily accessible.

Next, we consider three standard measures of liquidity: Amihud's (2002) illiquidity measure (AIM), Goyenko, Holden, and Trzcinka's (2009) effective tick measure, and Lesmond, Ogden, and Trzcinka's (1999) fraction of trading days with zero or missing returns. If, as we argue, EDGAR inclusion reduces information asymmetries among investors, we expect liquidity to improve. The estimates shown in columns 3 through 5 support this prediction. All three measures decline in the quarter of EDGAR inclusion and over the next four quarters, suggesting that liquidity improves. AIM, for example, drops by 4.9% from the pretreatment mean over the four quarters following EDGAR inclusion ($p = .005$).²²

Finally, if less costly, timelier, and more equal access to corporate disclosures reduces uncertainty about a firm's prospects, we expect volatility to fall. Column 6 supports this prediction: over the four quarters post-EDGAR inclusion, volatility falls by 2.7% from its pretreatment mean ($p = .037$).

²² In their Internet Appendix, Gao and Huang (2020) similarly find that EDGAR inclusion improves liquidity.

The results in Table 2 suggest that EDGAR inclusion is a meaningful shock to firms' information environments: investors respond by trading more, liquidity improves, and volatility declines. We next investigate how analysts respond to EDGAR inclusion.

2.2 Changes in analyst behavior around EDGAR inclusion

We examine analysts' responses to a firm's mandatory disclosures becoming universally available on both the extensive margin (does the analyst continue to cover the stock?) and the intensive margin (does she change her behavior?). We model responses both at the stock level (asking, e.g., how the number of analysts or forecast dispersion changes) and at the analyst-stock level (asking, e.g., how a given analyst changes her coverage of or forecasts for a given stock around EDGAR inclusion). In the remainder of this section, we present arguably causal evidence that analysts change their behavior in every dimension we consider. In Section 3, we explore possible reasons for why they do so.

2.2.1 Analyst coverage. A priori, the effect of joining EDGAR on analyst coverage is ambiguous. Universal access to corporate filings could reduce analysts' information production costs and so encourage an increase in coverage (Verrecchia 1982; Kim and Verrecchia 1994). On the other hand, cheaper and timelier access to corporate disclosures among investors could reduce the value of analysts' information production, inducing exit in the form of reduced coverage (Dugast and Foucault 2018).

Table 3 reports the results. In column 1, we see a significant decline in the number of analysts covering a firm once it joins EDGAR (down by 8.1% compared to before) and over the next four quarters ($p < .001$). In column 3, we see a significant decline in the likelihood that a *given* analyst continues to cover a *given* stock post-EDGAR. This likelihood falls by 3.1% in the treatment quarter ($p < .001$) and remains 4% lower over the subsequent four quarters ($p < .001$).²³

The internal validity of our DD analysis requires that treated and control firms would have experienced similar trends in coverage but for the EDGAR treatment. A common way to gauge the plausibility of the parallel trends assumption is to check for the absence of diverging trends before treatment. Figure 2 plots dynamic event-study DD estimates of the effects of EDGAR inclusion on coverage over our nine-quarter window, along with 95%

²³ Our result that analyst coverage declines post-EDGAR contrasts with Gao and Huang's (2020) result that coverage increases. The difference in results reflects differences in research design. Our emphasis on covariate balance and internal validity results in a sample that skews toward smaller firms, for which the effect of access to SEC filings is arguably larger than for the larger firms studied by Gao and Huang. We can replicate Gao and Huang's result if we simultaneously relax the requirement that controls be matched on size and use the SEC's preliminary dates for waves 1–4 (rather than Jan. 1994, when the NSF-funded online access started) and for waves 5–10 (rather than the SEC's subsequently revised, actual dates).

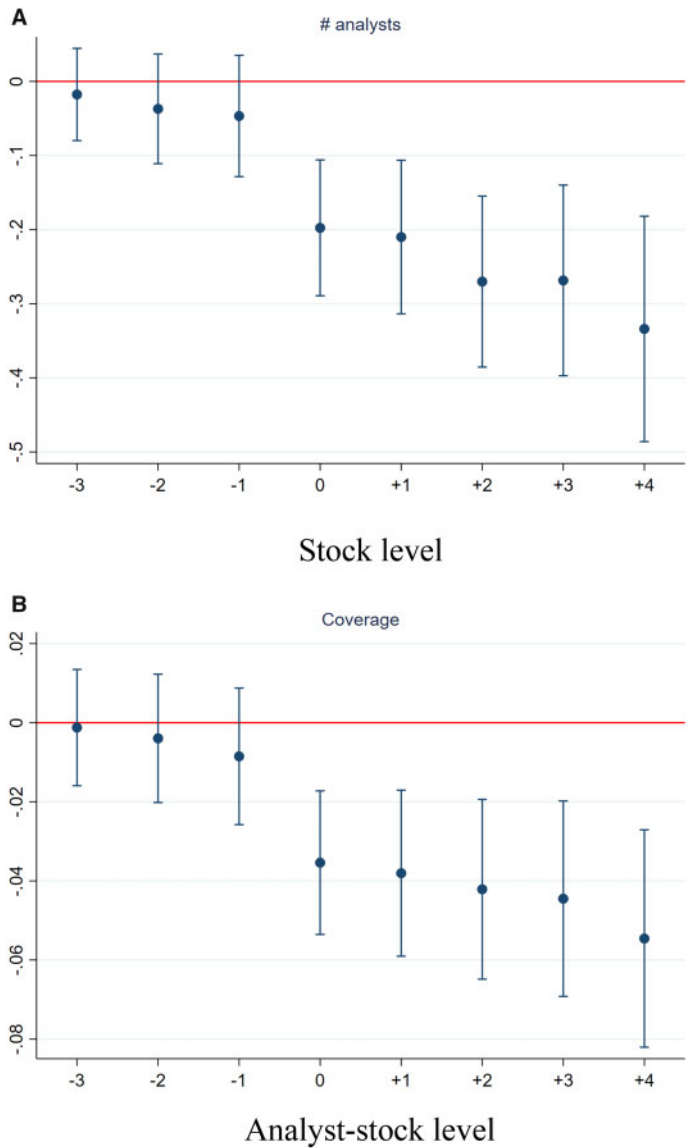


Figure 2
Testing for diverging pre-trends: Analyst coverage

The figure graphs difference-in-differences estimates of the effects of inclusion in EDGAR on analyst coverage. Treated firms are those included in EDGAR at time 0; control firms are nearest-neighbor propensity-score matched on equity market capitalization (in levels and logs), the number of analysts (in logs and lags), and the fiscal quarter using a 0.05 caliper. We include data from a nine fiscal quarter window centered on the fiscal quarter in which a treated firm's EDGAR inclusion takes place. All specifications are estimated using ordinary least squares (OLS) and include controls (the one-quarter lag of log market cap) and fixed effects for calendar-quarter, fiscal-quarter, and firm (or analyst-firm). The vertical lines represent 95% confidence intervals. For variable definitions and details of their construction, see Appendix B.

Table 3
Extensive margin: Analyst coverage

	Stock level: # analysts		Analyst-stock level: = 1 if provides coverage	
	(1)	(2)	(3)	(4)
Quarter of EDGAR inclusion	-0.170*** <i>0.033</i>	0.228 <i>0.144</i>	-0.031*** <i>0.008</i>	0.024 <i>0.037</i>
x disclosure quality (<i>DQ</i>)		-0.606*** <i>0.217</i>		-0.083 <i>0.055</i>
Next four quarters	-0.232*** <i>0.043</i>	0.148 <i>0.131</i>	-0.040*** <i>0.008</i>	0.050 <i>0.032</i>
x disclosure quality (<i>DQ</i>)		-0.577*** <i>0.194</i>		-0.135*** <i>0.046</i>
Controls?	Yes	Yes	Yes	Yes
Calendar quarter FE?	Yes	Yes	Yes	Yes
Fiscal quarter FE?	Yes	Yes	Yes	Yes
Firm FE?	Yes	Yes		
Analyst-firm FE?			Yes	Yes
R-squared (%)	89.9	89.9	35.0	35.0
No. of firms	4,032	4,032	2,290	2,290
No. of observations	33,924	33,924	135,260	135,260

The table reports difference-in-differences estimates of the effects of inclusion in EDGAR on analyst coverage at the stock level (using the quarterly count of analysts covering the stock) and at the analyst-stock level (using an indicator set equal to one if the analyst covers the stock in a quarter and zero otherwise). Treated firms are those included in EDGAR; control firms are nearest-neighbor propensity-score matched on equity market capitalization (in levels and logs), number of analysts (in logs and lags), and fiscal quarter using a 0.05 caliper. In columns 2 and 4, we interact treatment with the firm's disclosure quality (*DQ*) score, measured in the quarter before treatment. We include data from a nine-fiscal quarter window centered on the fiscal quarter in which a treated firm's EDGAR inclusion takes place. All specifications are estimated using OLS and include controls (the one-quarter lag of log market cap) and fixed effects for calendar-quarter, fiscal-quarter, and firm (or analyst-firm). For variable definitions and details of their construction, see Appendix B. Heteroscedasticity-consistent standard errors are shown in italics underneath the coefficient estimates. They are clustered at the firm level in columns 1 and 2 and double-clustered at the firm and analyst-quarter level in column 3. * $p < .1$; ** $p < .05$; *** $p < .01$.

confidence intervals. The figure confirms the absence of diverging pre-trends, both at the stock level and at the analyst-stock level: coverage is similar among treated and control firms in the quarters before EDGAR inclusion and then falls significantly among treated firms in the quarter they join EDGAR, without recovering over the next four quarters.

The absence of diverging pre-trends supports a causal interpretation of the patterns in Table 3 and Figure 2. EDGAR inclusion reduces analyst coverage, suggesting that mandatory disclosure and information production by analysts are substitutes to some extent. This finding is consistent with Dugast and Foucault (2018), whose model predicts that greater access to corporate filings reduces demand for fundamental analysis.

To investigate this crowding-out mechanism further, columns 2 and 4 interact treatment with Chen, Miao, and Shevlin's (2015) *DQ* measure (which captures the granularity of a firm's financial reports), evaluated in the quarter before treatment. The interaction effects are negative, both at the stock level and at the analyst-stock level, indicating that firms with the highest disclosure quality leave less room for analysts to add value once filings become universally available. If we interpret *DQ* as a proxy for the precision of the

public signal (as in Fischer and Stocken 2010), the results in Table 3 suggest that a more precise public signal, once freely and universally available, crowds out analyst information production.

2.2.2 Forecast optimism. Prior work has shown that analysts' earnings forecasts are, on average, overly optimistic and that strategic considerations (reflecting career concerns, compensation incentives, or a desire to stay on good terms with managers) may be at play. How does optimism (the scaled difference between forecast and realized earnings) change when investors gain free, timely, and equal access to mandatory disclosures?

Table 4 reports the results. At the stock level, average optimism falls for both short- and long-term forecasts in the quarter a firm joins EDGAR and remains at a significantly lower level over the next four quarters. Each DD estimate is statistically significant and economically sizeable. As an illustration, we find that joining EDGAR reduces average short-term optimism by 42%, from 0.008 in the pretreatment quarter to around 0.005 in the treatment quarter and the next four quarters.²⁴ At the analyst-stock level, we see that a given analyst issues forecasts that are less optimistic post-EDGAR than that same analyst's forecasts were for that stock in the previous four quarters. This reduction is statistically significant for both short- and long-term forecasts starting in the treatment quarter and does not revert back over the next four quarters ($p < .001$).²⁵ Figure 3 plots the corresponding dynamic DD estimates, confirming the absence of significantly diverging pre-trends as well as the persistence in the decline in optimism.

2.2.3 Inaccuracy. Less optimistic forecasts are not necessarily the same as more accurate forecasts. To assess accuracy requires measuring forecast errors, taking the absolute value of the difference between a forecast and realized earnings (appropriately scaled). We refer to these forecast errors as "inaccuracy," given that larger errors correspond to less accurate forecasts. Table 4 shows that inaccuracy falls following EDGAR inclusion and that forecasts remain significantly more accurate in the following four quarters. This is true for both short- and long-term forecasts. It is also true both for the average forecast made for a given firm and for a given analyst-firm pair. The improvements in accuracy are economically sizeable. As an illustration, we can observe that joining EDGAR reduces average short-term inaccuracy

²⁴ Our finding that optimism falls post-EDGAR is not driven by outliers. Table IA.1 in the Internet Appendix shows that the fraction of analysts whose forecasts exceed subsequent earnings (at the stock level) and the likelihood of an analyst's forecast exceeding subsequent earnings (at the analyst-stock level) fall significantly around EDGAR inclusion, by an economically large 4 to 8 percentage points, depending on specification.

²⁵ The fact that we find reductions in optimism both at the stock level and at the analyst-stock level rules out the possibility that optimism falls around EDGAR inclusion simply because the more optimistic analysts drop coverage. Indeed, as Table IA.2 in the Internet Appendix shows the tendency of analysts to drop coverage when a firm joins EDGAR is unrelated to their pre-EDGAR optimism. It is also unrelated to broker size.

Table 4
Analyst behavior: Forecast optimism and inaccuracy

	Optimism				Inaccuracy			
	Stock level		Analyst-stock level		Stock level		Analyst-stock level	
	Short-term (1)	Long-term (2)	Short-term (3)	Long-term (4)	Short-term (5)	Long-term (6)	Short-term (7)	Long-term (8)
Quarter of EDGAR inclusion	-0.336*** 0.121	-0.421 0.293	-0.502*** 0.144	-0.648*** 0.245	-0.269** 0.109	-0.354 0.270	-0.420*** 0.132	-0.463*** 0.221
Next four quarters	-0.377*** 0.112	-0.920** 0.385	-0.595*** 0.156	-1.091*** 0.321	-0.262** 0.102	-0.730** 0.364	-0.429*** 0.137	-0.778*** 0.277
Controls?	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Calendar quarter FE?	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Fiscal quarter FE?	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Firm FE?	Yes	Yes			Yes	Yes		
Analyst-firm FE?			Yes	Yes			Yes	Yes
R-squared (%)	41.5	58.2	57.0	67.5	49.2	64.5	61.7	71.3
No. of firms	2,251	2,832	2,251	2,832	2,251	2,832	2,251	2,832
No. of observations	10,857	17,980	22,010	60,640	10,857	17,980	22,010	60,640

The table reports difference-in-differences estimates of the effects of inclusion in EDGAR on forecast optimism and inaccuracy. Treated firms are those included in EDGAR; control firms are nearest-neighbor propensity-score matched on equity market capitalization (in levels and logs), number of analysts (in logs and lags), and fiscal quarter using a 0.05 caliper. We include data from a nine-fiscal quarter window centered on the fiscal quarter in which a treated firm's EDGAR inclusion takes place. All specifications are estimated using OLS and include controls (the one-quarter lag of log market cap) and fixed effects for calendar-quarter, fiscal-quarter, and firm (or analyst-firm). For variable definitions and details of their construction, see Appendix B. The coefficients are multiplied by 100 for ease of exposition. Heteroscedasticity-consistent standard errors are shown in italics underneath the coefficient estimates. They are clustered at the firm level in columns 1–2 and 5–6 and double-clustered at the firm and analyst-quarter level in columns 3–4 and 7–8. * $p < .1$; ** $p < .05$; *** $p < .01$.

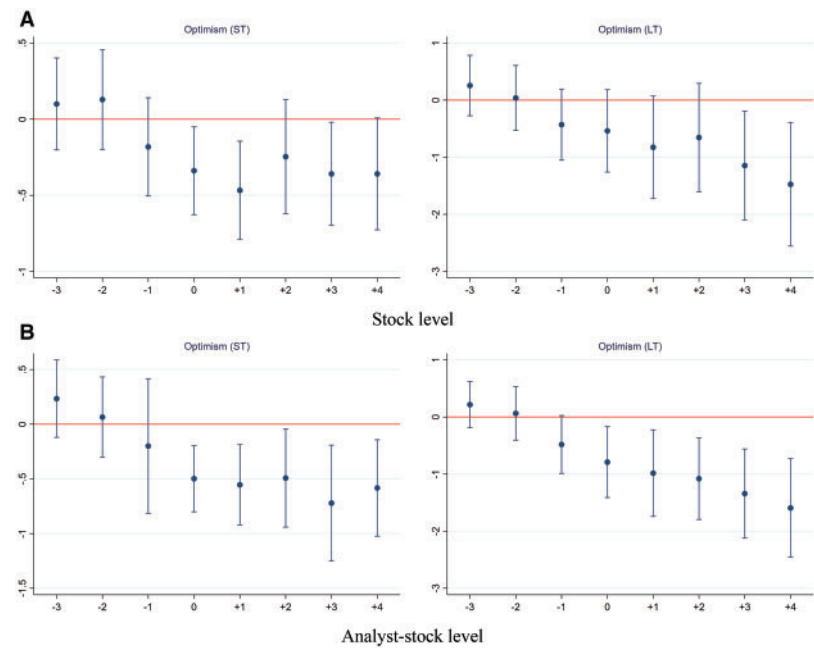


Figure 3
Testing for diverging pre-trends: Forecast optimism

The figure graphs difference-in-differences estimates of the effects of inclusion in EDGAR on forecast optimism. Treated firms are those included in EDGAR at time 0; control firms are nearest-neighbor propensity-score matched on equity market capitalization (in levels and logs), number of analysts (in logs and lags), and fiscal quarter using a 0.05 caliper. We include data from a nine-fiscal quarter window centered on the fiscal quarter in which a treated firm's EDGAR inclusion takes place. All specifications are estimated using OLS and include controls (the one-quarter lag of log market cap) and fixed effects for calendar-quarter, fiscal-quarter, and firm (or analyst-firm). The vertical lines represent 95% confidence intervals. For variable definitions and details of their construction, see Appendix B.

by 22%, from 0.012 in the pretreatment quarter to 0.009 in the treatment quarter, without reverting back over the next four quarters. Figure 4 plots the corresponding dynamic DD estimates. There is no evidence of significantly diverging pre-trends, consistent with the parallel trends assumption required for identification.

2.2.4 Quantile regressions. Figures 5 and 6 graph estimates from quantile DD regressions for optimism and inaccuracy, respectively, at (a) the stock level and (b) the analyst-stock level, separately for short- and long-term forecasts. In all eight graphs, there is a pronounced negative slope, such that the reduction in optimism or inaccuracy is larger the larger the initial level of optimism or inaccuracy. This is true both within stock and within an analyst-stock pair. The variation in the economic magnitude of the effects across deciles is large. As an illustration, while average short-term optimism in Figure 5, panel A, falls by an average of 30% from the pre-EDGAR mean in the decile of stocks with

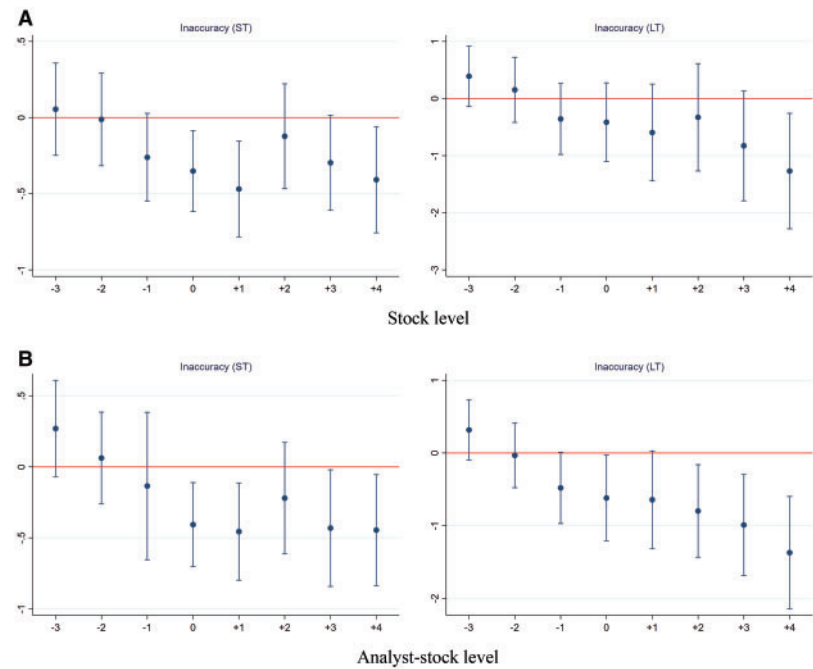


Figure 4
Testing for diverging pre-trends: Forecast inaccuracy

The figure graphs difference-in-differences estimates of the effects of inclusion in EDGAR on forecast inaccuracy. Treated firms are those included in EDGAR at time 0; control firms are nearest-neighbor propensity-score matched on equity market capitalization (in levels and logs), number of analysts (in logs and lags), and fiscal quarter using a 0.05 caliper. We include data from a nine-fiscal quarter window centered on the fiscal quarter in which a treated firm's EDGAR inclusion takes place. All specifications are estimated using OLS and include controls (the one-quarter lag of log market cap) and fixed effects for calendar-quarter, fiscal-quarter, and firm (or analyst-firm). The vertical lines represent 95% confidence intervals. For variable definitions and details of their construction, see Appendix B.

the lowest initial optimism ($p=.001$), it falls by 71% in the decile of stocks with the highest initial optimism ($p<.001$).

2.2.5 Auxiliary tests. Table IA.3 in the Internet Appendix reports two auxiliary tests that exploit variation across EDGAR joiners in two disclosure characteristics: how aggressively the firm used discretionary accruals before joining EDGAR ($|DA(Jones)_{-1}|$) and a proxy for firms with more extensive disclosures (*reporting richness*₋₁). Each is measured in the quarter before EDGAR inclusion and then interacted with our treatment indicators in a standard triple-diff model. Bradshaw, Richardson, and Sloan (2001) show that analysts “do not alert investors to the future earnings problems associated with high accruals,” either because they “lack the necessary sophistication” or because they “collude with management to inflate expectations of future earnings by inflating current accruals, current earnings, and forecasts of

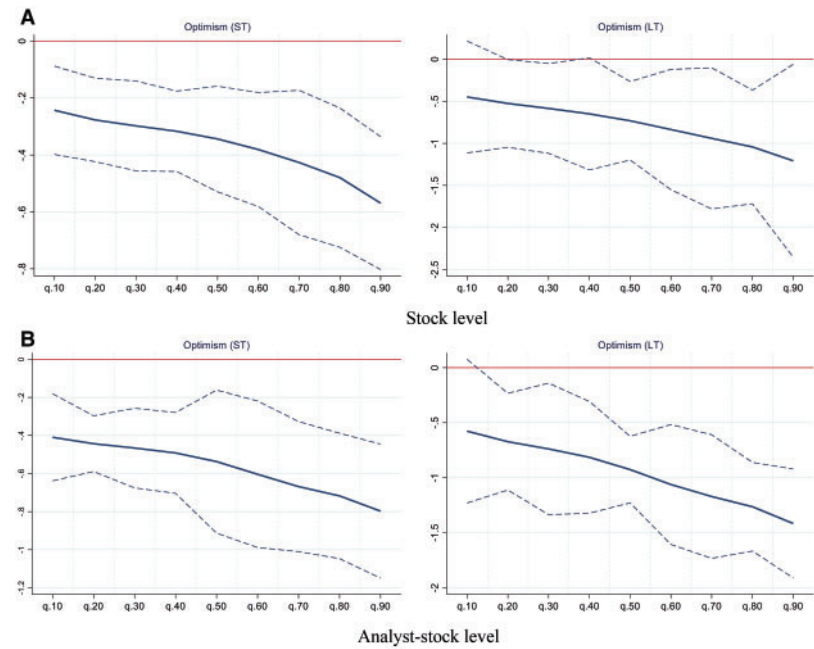


Figure 5
Quantile regressions: Forecast optimism
The figure graphs quantile-regression DD estimates of the effects of inclusion in EDGAR on forecast optimism. Treated firms are those included in EDGAR; control firms are nearest-neighbor propensity-score matched on equity market capitalization (in levels and logs), the number of analysts (in logs and lags), and the fiscal quarter using a 0.05 caliper. All specifications are estimated using OLS and include controls (the one-quarter lag of log market cap) and fixed effects for calendar-quarter, fiscal-quarter, and firm (or analyst-firm). The dashed lines represent 95% confidence intervals. For variable definitions and details of their construction, see Appendix B.

future earnings.” Whether investors are more “sophisticated” than analysts is an open question, but if analysts “collude” with management, EDGAR should help investors police this form of strategic analyst behavior, by giving investors access to disaggregate information about accruals that is available in corporate filings, but not in earnings releases. Table IA.3 shows that analysts moderate both optimism and inaccuracy by significantly more, the more aggressively the firm used discretionary accruals before joining EDGAR. In other words, it is in the stocks with the greatest scope for strategic behavior that analysts moderate their behavior the most when corporate filings become freely available via EDGAR. This finding is consistent with analysts behaving strategically.

In the cross-section of firms, analysts should moderate their behavior more in firms whose EDGAR filings are likely to contain more extensive disclosure: the more information a firm discloses, the better able investors are to validate the accuracy of analyst forecasts. Table IA.3 tests this prediction using a measure of reporting richness borrowed from Coles, Daniel, and Naveen (2008) based

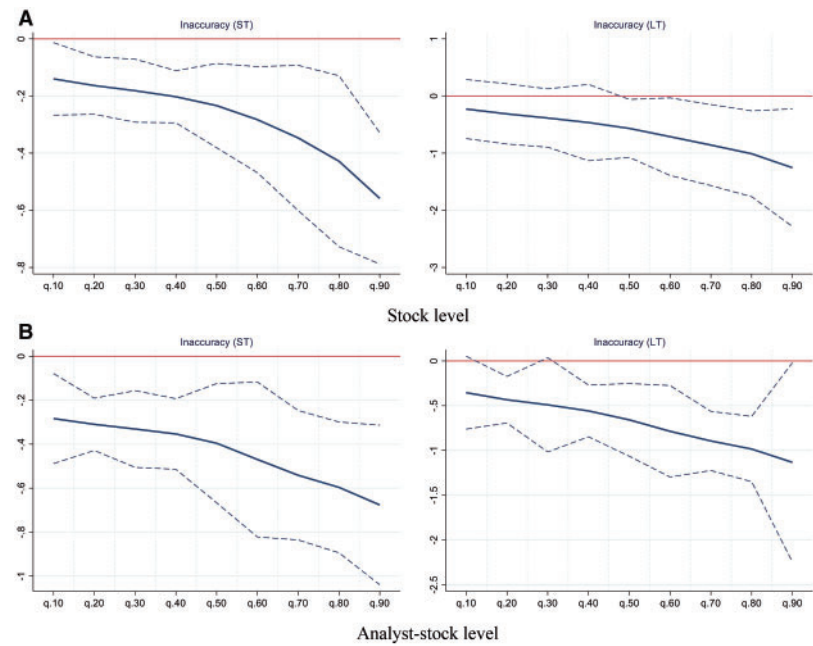


Figure 6
Quantile regressions: Forecast inaccuracy

The figure graphs quantile-regression DD estimates of the effects of inclusion in EDGAR on forecast inaccuracy. Treated firms are those included in EDGAR; control firms are nearest-neighbor propensity-score matched on equity market capitalization (in levels and logs), the number of analysts (in logs and lags), and the fiscal quarter using a 0.05 caliper. All specifications are estimated using OLS and include controls (the one-quarter lag of log market cap) and fixed effects for calendar-quarter, fiscal-quarter, and firm (or analyst-firm). The dashed lines indicate 95% confidence intervals. For variable definitions and details of their construction, see Appendix B.

on segment reporting, firm size, and leverage.²⁶ Consistent with the prediction, we find that analysts moderate their forecast behavior significantly more in firms with more extensive disclosures.

2.2.6 Robustness tests. The observed reductions in optimism and inaccuracy in Table 4 could be mechanical to the extent that EDGAR inclusion might affect forecast timing. Richardson, Teoh, and Wysocki (2004) show that analysts start with optimistic forecasts and then “walk down” their forecasts closer to the earnings announcement, perhaps to make it easier for companies to beat their forecasts. If EDGAR inclusion induces analysts to make later forecasts, we would expect to see less optimistic and more accurate forecasts post-EDGAR. However, Table IA.4 in the Internet Appendix shows that forecast timing is unchanged around EDGAR inclusion. Moreover, Table IA.5 shows that the

²⁶ Firms reporting results by segment, larger firms, and firms that borrow more heavily have more extensive disclosures (Leuz and Verrecchia 2000).

reductions in optimism and inaccuracy reported in Table 4 are little different in economic magnitude and statistical significance if we measure optimism and inaccuracy using only an analyst's last forecast in a fiscal quarter.

2.2.7 Informativeness of analyst forecasts and recommendations. Analyst forecasts move stock prices when they are seen as revealing new information in the eyes of the marginal investor. We predict that there is less scope for analysts to move stock prices when the marginal investor is given free, timely, and equal access to corporate disclosures.

We measure informativeness as the price impact that can be attributed to forecasts, using either Lehavy, Li, and Merkley's (2011) unsigned measure or Park and Stice's (2000) revision response coefficient. In either case, Table 5 shows that investors view forecasts as less informative once a stock joins EDGAR. For the average treated stock, informativeness using the unsigned measure declines by 11%, from 0.071 in the quarter before treatment to 0.063 in the treatment quarter, without reverting back over the next four quarters ($p < .001$). The results are similar for the revision response coefficient. Informativeness declines even though the average analyst forecast has become both less biased and less noisy (Table 4), implying that the improvements in firms' information environments that result from corporate filings becoming publicly available outweighs the beneficial reduction in forecast bias and inaccuracy.^{27,28} Like our earlier results showing a reduction in coverage, we interpret these findings to suggest that mandatory disclosure and information production by analysts are substitutes in this setting.

Recommendation changes, shown in column 3, also move prices by significantly less post-EDGAR ($p < .001$), suggesting that investors become more skeptical of analysts' recommendations. This is driven by a change in investors' responses to recommendation upgrades (column 4).

2.2.8 Forecast dispersion. We can expect forecast dispersion to decline post-EDGAR for at least three reasons. First, reduced optimism and increased accuracy mechanically reduce dispersion. Second, to the extent that EDGAR inclusion improves some analysts' access to mandatory disclosures (e.g., those at smaller brokerage houses that could not justify the expense of a subscription to data vendors, such as Mead Data Central or Dialog), we expect information asymmetries, and hence dispersion, among analysts to decline.

²⁷ During the EDGAR inclusion quarter, a small number of forecast revisions predate the firm's first EDGAR filings. Their inclusion in Table 5 is conservative, in that they attenuate the estimated reduction in forecast informativeness as investors do not yet have access to EDGAR filings. Purging the informativeness measure of these early forecasts increases the estimated reduction in informativeness by around 25% in absolute terms, consistent with the results in Table 5 being conservative.

²⁸ These results contrast with Gao and Huang's (2020) finding that analyst forecasts become more informative post-EDGAR. We can replicate Gao and Huang's finding if we make the same changes that are necessary to replicate their finding that coverage increases (see footnote 23).

Table 5
Analyst behavior: Informativeness, dispersion, and boldness

Infor- mative- ness	coefficient (2)	Recommendation changes			Dispersion		Boldness	
		Pooled (3)	Up (4)	Down (5)	Short-term (6)	Long-term (7)	Short-term (8)	Long-term (9)
Quarter of EDGAR inclusion	-0.765*** 0.192	-0.854** 0.416	-1.119** 0.484	-0.615 0.484	-0.004 0.011	-0.013 0.040	0.034 0.049	0.066 0.052
Next four quarters	-0.965*** 0.186	-0.642* 0.342	-1.242** 0.523	-0.670 0.519	-0.034** 0.013	-0.139*** 0.048	-0.083* 0.049	-0.136** 0.055
Controls?	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Calendar quarter FE?	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Fiscal quarter FE?	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Firm FE?	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Analyst-firm FE?							Yes	Yes
R-squared (%)	57.6	15.8	37.1	40.9	46.2	67.7	66.4	69.5
No. of firms	2,836	2,556	2,079	1,733	1,927	1,665	2,298	1,954
No. of observations	18,394	14,148	6,889	4,314	4,967	9,302	16,681	35,313

The table reports difference-in-differences estimates of the effects of inclusion in EDGAR on two measures of forecast informativeness, the price impact of recommendation changes, forecast dispersion, and forecast boldness. Treated firms are those included in EDGAR; control firms are nearest-neighbor propensity-score matched on equity market capitalization (in levels and logs), number of analysts (in logs and lags), and fiscal quarter using a 0.05 caliper. We include data from a nine-fiscal quarter window centered on the fiscal quarter in which a treated firm's EDGAR inclusion takes place. All specifications are estimated using OLS and include controls (the one-quarter lag of log market cap) and fixed effects for calendar-quarter, fiscal-quarter, and firm (or analyst-firm). For variable definitions and details of their construction, see Appendix B. Except in column 2, the coefficients are multiplied by 100 for ease of exposition. Heteroscedasticity-consistent standard errors are shown in italics underneath the coefficient estimates. They are clustered at the firm level in columns 1 through 5 and double-clustered at the firm and analyst-quarter level in columns 8 and 9. * $p < .1$; ** $p < .05$; *** $p < .01$.

Third, EDGAR inclusion may increase analysts' incentives to herd rather than stand out from the crowd. Our working hypothesis is that universal access to corporate disclosures makes it easier for investors to evaluate an analyst's forecast performance. If so, easier ex post scrutiny could discourage the kinds of long-shot (or bold) forecasts that could hurt an analyst's career if later proven wrong.²⁹

Table 5, columns 4 and 5, confirms our prediction: dispersion in both short- and long-term forecasts declines significantly, beginning in the quarter after EDGAR inclusion ($p < .01$). The reductions are economically meaningful, averaging 11% for short-term dispersion and 15% for long-term dispersion.

Columns 6 and 7 use a standard analyst-stock level measure of boldness borrowed from Hong, Kubik, and Solomon (2000), showing that a given analyst makes significantly less bold forecasts for a given firm after the firm joins EDGAR, all else equal. Economically, boldness falls by 21% for short-term forecasts ($p = .092$) and by 17% for long-term forecasts ($p = .014$).

2.2.9 Analyst recommendations. Our findings so far suggest that analysts change their earnings forecasts systematically around a firm's inclusion in EDGAR: forecasts become significantly less optimistic and more accurate. Table 6 considers how analysts' buy/sell recommendations change around EDGAR. Interestingly, we find that analysts *increase* the strength of their recommendations significantly following EDGAR inclusion, both as a group (i.e., at the stock level) and individually (i.e., at the analyst/stock level). The point estimates in column 2 suggest that 1 in 10 analysts upgrade the stock by one notch (say, from buy to strong buy) when a firm joins EDGAR ($p = .004$) and retains that upgraded recommendation over the following year ($p < .001$).³⁰

Tables 4, 5, and 6 suggest that as analysts become more conservative in their forecasting behavior when a stock joins EDGAR, their recommendations simultaneously become more bullish, while investors become more skeptical. We return to an interpretation of this set of findings in Section 3.

2.3 Identification concerns

DD models like ours make certain identifying assumptions, which need to be satisfied for DD estimates to be interpreted as causal. First, treatment must be randomly assigned, or else systematic unobserved differences between

²⁹ For example, issuing bold forecasts runs the risk of being labeled "low ability" if investors believe high-ability analysts receive correlated information (Scharfstein and Stein 1990; Prendergast and Stole 1996).

³⁰ Column 3 reports the results of a falsification test, using a separate sample of recommendations by Value Line analysts. Because Value Line does not offer brokerage or underwriting services, Value Line analysts are considered free from the conflicts of interest that the literature views as potential sources of strategic analyst behavior, unlike the analysts considered in our baseline sample (Eames and Glover 2003). We find no evidence that Value Line analysts change their recommendations around EDGAR inclusion. (We do not have access to Value Line earnings forecasts and so cannot test how those might change around EDGAR inclusion.)

Table 6
Analyst Behavior: Recommendation Strength

	Recommendation strength		
	Stock level (1)	Analyst-stock level (2)	Stock-level falsification test (3)
Quarter of EDGAR inclusion	0.055* <i>0.031</i>	0.102*** <i>0.036</i>	-0.201 <i>0.205</i>
Next four quarters	0.078** <i>0.034</i>	0.124*** <i>0.032</i>	-0.333 <i>0.222</i>
Controls?	Yes	Yes	Yes
Calendar quarter FE?	Yes	Yes	Yes
Fiscal quarter FE?	Yes	Yes	Yes
Firm FE?	Yes		Yes
Analyst-firm FE?		Yes	
R-squared (%)	56.7	69.4	61.8
No. of firms	2,766	2,725	554
No. of observations	18,841	27,676	3,700

The table reports difference-in-differences estimates of the effects of inclusion in EDGAR on the strength of analysts' stock recommendations. Columns 1 and 2 focus on recommendations issued by the sellside analysts studied in Tables 3, 4, and 5. Column 3 reports a falsification test using recommendations issued by Value Line analysts. Value Line recommendations are obtained from Zack's Investment Research and are available only for calendar years 1993 and 1994. We reverse-score the five-point recommendation scale used in our data sources such that a 5 corresponds to a strong buy and a 1 corresponds to a strong sell. Treated firms are those included in EDGAR; control firms are nearest-neighbor propensity-score matched on equity market capitalization (in levels and logs), number of analysts (in logs and lags), and fiscal quarter using a 0.05 caliper. We include data from a nine-fiscal quarter window centered on the fiscal quarter in which a treated firm's EDGAR inclusion takes place. All specifications are estimated using OLS and include controls (the one-quarter lag of log market cap) and fixed effects for calendar-quarter, fiscal-quarter, and firm (or analyst-firm). For variable definitions and details of their construction, see Appendix B. Heteroscedasticity-consistent standard errors are shown in italics underneath the coefficient estimates. They are clustered at the firm level in column 1 and double-clustered at the firm and analyst-quarter level in column 2. ** $p < .05$; *** $p < .01$.

treated and controls could cause posttreatment differences between treated and controls that are nothing to do with the treatment. This assumption is arguably satisfied given the way the SEC implemented the transition to EDGAR.

Second, and closely related to random assignment, the difference between treated and controls must be constant over time in the absence of treatment. Conditional random assignment, goes a long way to ensuring that this parallel trends assumption is likely to hold, by eliminating concerns that treated and controls differ systematically on unobservables that could cause differences in posttreatment trends to emerge. The fact that we fail to find diverging pre-trends further supports the parallel trends assumption.

Third, treatment must not coincide with other events that affect the treated and controls differently. Conditional random assignment coupled with the staggered rollout of EDGAR, greatly reduces the scope for violations of this assumption: random assignment ensures that treated and controls are not plausibly differentially sensitive to unobserved contemporaneous shocks, and staggering ensures that firms are treated at different times on a schedule that is unlikely to coincide with unobserved shocks.

Fourth, treatment must be unexpected, or else treated firms (and potentially controls) could adjust to treatment prior to treatment in ways that could

confound the estimated treatment effect. For example, analysts might change their behavior before a stock joins EDGAR, knowing that universal access to corporate filings will eventually allow more investors to verify analyst reports in embarrassing ways. While EDGAR itself was not a surprise, two features of its implementation arguably were. The first is that EDGAR was not, when it was announced, intended to provide universal access to corporate filings. Instead, the SEC announced EDGAR as an electronic *filing* system. Only once the NSF funded NYU's attempts to put EDGAR online from January 17, 1994, did EDGAR become an electronic *access* system. This means that firms in the first four waves arguably were not expected to have their filings accessible online. Importantly, our results are robust to using only the first four waves (see Table IA.6). The second feature is that the SEC announced assignments to waves 5 through 10 only in December 1994. Thus, firms in these waves did not know their EDGAR join dates until a few months before joining.

Finally, the effects of treatment must be confined to the treated and not spill over to controls, or else interactions between treated and controls could lead to bias. In our setting, this stable unit treatment value assumption (SUTVA) would be violated if analyst k changed her forecasting behavior in quarter t not just for those stocks i_k that join EDGAR at t but also for the other stocks $\neg i_k$ she covers that will join EDGAR at a future time and so serve as our controls.³¹ Of course, violations of SUTVA work against us finding any effect of EDGAR inclusion on analyst behavior: if analysts did change their behavior for both treated and controls, our DD estimates would be attenuated toward zero.

A small change to our empirical design allows us to test for violations of SUTVA. So far, we have coded as the treated unit either stock i or an analyst-stock pair i_k . Now, we code analyst k as being treated from the time one or more of her stocks first joins EDGAR and ask how her forecasting behavior differs between those stocks joining EDGAR and those that have not yet joined EDGAR. In contrast to our baseline models, we thus hold the analyst constant. If the analyst changes her behavior for all her stocks when only some are included in EDGAR, the coefficients will be zero. The results, reported in Table IA.7 in the Internet Appendix, show that a given analyst reduces her optimism and inaccuracy by significantly more in stocks joining EDGAR than in those that have not yet joined EDGAR. Moreover, the estimates in Table IA.7 are economically quite close to the baseline estimates in Table 4, suggesting that potential violations of SUTVA have little material effect in our setting.³²

³¹ One reason the analyst might adjust her behavior simultaneously for current treated and future treated stocks is that she anticipates that even stocks that have not yet joined EDGAR will eventually join EDGAR, which will change the nature of the game she plays with investors and firms.

³² To illustrate, Table 4 shows that analysts reduce short-term optimism for stocks joining EDGAR by an average of 0.595 relative to future-treated control stocks *covered by themselves or other analysts*, while in Table IA.7, they reduce short-term optimism by an average of 0.706 relative to future-treated control stocks *covered by themselves*.

2.4 Impact on investors

So far, we have reported arguably causal evidence that analysts change their behavior around EDGAR inclusion, an event that investors appear to regard as sufficiently important so that trading volume, liquidity, and volatility all change in response. Before we consider possible reasons for why analysts change their behavior, we briefly consider how EDGAR affects investors.

The finding that coverage falls may be detrimental to investors if it reduces information production about a stock. The finding that forecast bias and errors both fall may on net be beneficial to investors, to the extent that the task of debiasing signals received from analysts becomes easier as a result. The finding that forecast dispersion falls could have the beneficial effect of reducing disagreement among investors, which in turn could make a stock less prone to crash risk (Chang et al. 2022). The finding that analysts make fewer bold forecasts could either harm investors (if it means that fewer outlier signals are incorporated in prices) or benefit them (if bold forecasts simply add noise to the consensus).

A summary measure of investor welfare eludes the literature. Instead, we consider what happens to standard measures of the net precision of the signals available to investors. The finding that forecasts carry less information post-EDGAR inclusion suggests the possibility that investors have become better informed. After all, investors combine signals from analyst reports and from firms' now more easily accessible mandatory disclosures to guide their trading decisions and their response to new information.

Table 7 focuses on three measures of how investors respond to the new information contained in earnings announcements. If EDGAR improves the net precision of the conditioning information investors have access to, investors should respond in a more muted way to earnings announcements than before. The three measures we use are the volatility of returns and volume of trading in a 3-day window around a firm's quarterly earnings announcement, and the speed with which stock prices adjust to the announcement. In each case, we find the expected attenuation in investor response. Earnings announcements are associated with significantly lower volatility ($p = .080$) and reduced trading ($p = .002$) after a firm joins EDGAR than before, all else equal, and stock prices converge significantly faster to the earnings news ($p = .043$), in the sense that prices change by less in absolute terms when earnings are announced (Heflin, Subramanyam, and Zhang 2003).

Overall, we interpret the results in Table 7 as suggesting that EDGAR inclusion improves the net precision of the signals investors base their trading decisions on.

3. Why Do Analysts Change Their Behavior?

We next investigate what we view as the two main competing explanations for why analyst behavior changes around EDGAR inclusion. First, EDGAR

Table 7
Effects on investor: Net precision

	Volatility at earnings announcement (1)	Volume at earnings announcement (2)	Price convergence (3)
Quarter of EDGAR inclusion	0.007 <i>0.017</i>	−0.052 <i>0.068</i>	−0.002 <i>0.003</i>
Next four quarters	−0.030* <i>0.017</i>	−0.283*** <i>0.091</i>	−0.007** <i>0.003</i>
Controls?	Yes	Yes	Yes
Calendar quarter FE?	Yes	Yes	Yes
Fiscal quarter FE?	Yes	Yes	Yes
Firm FE?	Yes	Yes	Yes
R-squared (%)	44.0	69.6	46.0
No. of firms	3,610	3,819	3,629
No. of firm-quarters	23,715	28,121	11,449

The table reports difference-in-differences estimates of the effects of inclusion in EDGAR on measures of the net precision of investors' information sets: volatility at earnings announcements, trading volume at earnings announcements, and price convergence. Treated firms are those included in EDGAR; control firms are nearest-neighbor propensity-score matched on equity market capitalization (in levels and logs), number of analysts (in logs and lags), and fiscal quarter using a 0.05 caliper. We include data from a nine-fiscal quarter window centered on the fiscal quarter in which a treated firm's EDGAR inclusion takes place. All specifications are estimated using OLS and include controls (the one-quarter lag of log market cap) and fixed effects for calendar-quarter, fiscal-quarter, and firm. For variable definitions and details of their construction, see Appendix B. Heteroscedasticity-consistent standard errors clustered at the firm level are shown in italics underneath the coefficient estimates. ** $p < .05$; *** $p < .01$.

inclusion could affect analysts' strategic behavior (the strategic-analyst channel). Second, EDGAR inclusion could reduce analysts' cost of information acquisition and thereby improve their information production ability (the broker channel). By way of preview, we find no evidence to support the broker channel and consistent evidence to support the strategic-analyst channel.³³

3.1 Strategic-analyst channel

We first present tests of the strategic-analyst channel that exploit heterogeneity in analysts' exposure to changes in investors' ability to verify their reports ex post and in analysts' incentives to strategically bias their forecasts ex ante. If free, timely, and equal access to corporate filings allows investors to more easily verify analyst reports (and assuming investors can punish analysts for issuing biased and inaccurate reports by, for example, reducing their trading intensity or moving their brokerage account), we expect EDGAR inclusion to raise the reputation cost to analysts of strategically misreporting their signal. Our tests reveal that it is the analysts who face the greatest change in the likelihood of detection and who have the greatest incentives to behave strategically pre-EDGAR who moderate their behavior the most post-EDGAR.

³³ We find no evidence consistent with a third possibility, namely, that EDGAR inclusion could cause changes in firms' fundamentals or disclosure policies to which analysts respond mechanically by changing their forecasts. See Section IA.1 and Table IA.8 in the Internet Appendix.

The institutional reality in the pre-EDGAR era, when commercial access to corporate filings was prohibitively costly for all but the largest market participants, implies that retail investors experienced a relatively larger reduction in the cost of accessing corporate filings (and so in verification costs), courtesy of EDGAR, than did large institutional investors (for many of which EDGAR made little difference in terms of access to corporate filings). We thus predict that analysts who serve retail clients should moderate their behavior by more when a stock joins EDGAR than analysts who serve institutional clients.³⁴

We measure a brokerage firm's retail focus as the share of its registered representatives who are licensed to provide advice to retail clients, using data gathered from Securities Industry Association yearbooks and measured as of the quarter before a firm joins EDGAR. The results from triple-diff regressions, reported in Table 8, support our prediction. The triple interaction *Treated* $\times$ *Post* $\times$ *Retail focus* is negative and statistically significant in all four specifications, confirming that the post-EDGAR reduction in optimism and inaccuracy is larger the greater a broker's focus on retail clients, that is, the more of an analyst's clients gain access to corporate filings courtesy of EDGAR. To illustrate the economic importance of retail investors, compare an analyst working at a retail-only brokerage firm to an analyst working at an institution-only broker. The coefficient estimates in Table 8 imply that the retail analyst reduces her short-term optimism by twice as much as the institutional analyst and her long-term optimism by 125% more than the institutional analyst, and that she improves her short-term accuracy by 74% more and her long-term accuracy by 132% more than the institutional analyst, all else equal. In short, a focus on retail investors has an economically large effect on the changes in analyst behavior we document around EDGAR inclusion.

The remainder of Table 8 investigates cross-analyst variation in two variables the literature associates with strategic analyst behavior. The first proxies for reputation. Some analysts have more to lose than others as investors' verification costs decline. In particular, we expect "star" analysts to moderate their behavior by more when a stock joins EDGAR than nonrated analysts.³⁵ The results support this prediction. The triple interaction *Treated* $\times$ *Post* $\times$ *Star analyst* is negative and significant, whether we look at optimism or

³⁴ This prediction implicitly assumes that brokerage firms care sufficiently about their retail business for their analysts to moderate their behavior post-EDGAR. We have good reason to expect that brokerage firms did. In the 1990s, retail investors owned almost half of the U.S. stock market directly (Barber and Odean 2000) and traded nearly as actively as mutual funds (Barber and Odean 2000; Kacperczyk, Sialm, and Zheng 2008), while retail investors generated considerably more revenue for brokerage firms because they paid much higher commission rates on their trades than institutional investors (Odean 1999; Keim and Madhavan 1997). Reflecting this institutional reality, large institutionally focused brokerage firms acquired retail-focused brokerage firms throughout the 1990s. Well-known examples include Morgan Stanley (which acquired Dean Witter), Salomon Brothers (which acquired Smith Barney), and UBS (which acquired PaineWebber).

³⁵ Stickel (1992) and Fang and Yasuda (2009) find that star analysts have more reputational capital at stake. By one estimate, the compensation of star analysts is 61% higher on average than that of their peers (Groysberg, Healy, and Maber 2011).

Table 8
Triple-difference estimates of the effects of inclusion in EDGAR on forecast optimism

	Optimism					
	Short-term			Long-term		
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Treated × Post</i>	−0.505*** <i>0.154</i>	−0.539*** <i>0.148</i>	−0.606*** <i>0.161</i>	−0.878*** <i>0.289</i>	−0.976*** <i>0.295</i>	−1.071*** <i>0.309</i>
<i>Treated × Post × Retail focus</i>	−0.005** <i>0.002</i>			−0.011*** <i>0.004</i>		
<i>Treated × Post × Star analyst</i>		−0.670* <i>0.372</i>			−1.051* <i>0.547</i>	
<i>Treated × Post × Affiliated analyst</i>			−0.548** <i>0.271</i>			−0.826 <i>0.525</i>
Controls?	Yes	Yes	Yes	Yes	Yes	Yes
Calendar quarter FE?	Yes	Yes	Yes	Yes	Yes	Yes
Fiscal quarter FE?	Yes	Yes	Yes	Yes	Yes	Yes
Analyst-firm FE?	Yes	Yes	Yes	Yes	Yes	Yes
<i>R-squared (%)</i>	54.2	53.5	57.0	65.2	63.9	67.5
<i>No. of firms</i>	2,062	2,019	2,251	2,606	2,538	2,832
<i>No. of observations</i>	18,247	17,672	22,010	50,639	48,070	60,640

For the corresponding difference-in-differences models, see the analyst-stock level models in Table 4. We interact treatment with three measures of an analyst's incentives to engage in strategic behavior, each measured in the quarter before EDGAR inclusion, so not time varying: how focused on retail investors the analyst's brokerage house is, whether the analyst is ranked as a "star," and whether the analyst is "affiliated" in the sense of working for a brokerage house that has an investment banking relationship with the firm that is joining EDGAR. (See Table IA.9 in the Internet Appendix for summary statistics.) *Post* equals one in the quarter of joining EDGAR and the next four quarters. Since the triple-interaction variables are measured in the quarter before treatment, they do not vary within firm and their levels effects are absorbed by the firm effects. All specifications are estimated using OLS and include controls (the one-quarter lag of log market cap) and fixed effects for calendar-quarter, fiscal-quarter, and analyst-firm. For variable definitions and details of their construction, see Appendix B. The coefficients are multiplied by 100 for ease of exposition. Heteroscedasticity-consistent standard errors double-clustered at firm and analyst-quarter levels are shown in italics underneath the coefficient estimates. * $p < .1$; ** $p < .05$; *** $p < .01$. (Continued)

inaccuracy and for both short- and long-term forecasts. This implies that stars reduce optimism and inaccuracy by significantly more when a stock joins EDGAR than do nonrated analysts (who, it is worth noting, also reduce optimism and inaccuracy significantly).

The second proxy seeks to capture a much-debated source of distorted incentives: conflicts of interest stemming from a broker's desire to keep its corporate clients happy (Michaely and Womack 1999; Ljungqvist, Marston, and Wilhelm 2006; Ljungqvist et al. 2007).³⁶ To capture such conflicts, we code as "affiliated" those analysts whose brokerage house provided debt or equity underwriting services to the focal firm in the 3 years before joining EDGAR. If EDGAR inclusion moderates strategic analyst

³⁶ Alleged conflicts of interest between research and investment banking were the stated reason for the 2003 "Global Settlement" between the SEC, FINRA, the New York Stock Exchange, and 10 large investment banks, requiring structural separation of research and investment banking. It is also a key motivation for the parts of the European Union's MiFID II Directive that "unbundle" the provision of research by investment banks.

Table 8
Triple-difference estimates of the effects of inclusion in EDGAR on forecast optimism (Continued)

	Inaccuracy					
	Short-term			Long-term		
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Treated × Post</i>	−0.408*** <i>0.140</i>	−0.398*** <i>0.135</i>	−0.484*** <i>0.145</i>	−0.604** <i>0.260</i>	−0.680*** <i>0.259</i>	−0.719*** <i>0.269</i>
<i>Treated × Post × Retail focus</i>	−0.003* <i>0.002</i>			−0.008** <i>0.004</i>		
<i>Treated × Post × Star analyst</i>		−0.668** <i>0.317</i>			−0.828* <i>0.494</i>	
<i>Treated × Post × Affiliated analyst</i>			−0.261 <i>0.242</i>			−0.904* <i>0.490</i>
Controls?	Yes	Yes	Yes	Yes	Yes	Yes
Calendar quarter FE?	Yes	Yes	Yes	Yes	Yes	Yes
Fiscal quarter FE?	Yes	Yes	Yes	Yes	Yes	Yes
Analyst-firm FE?	Yes	Yes	Yes	Yes	Yes	Yes
R-squared (%)	59.5	58.9	61.7	69.1	67.9	71.3
No. of firms	2,062	2,019	2,251	2,606	2,538	2,832
No. of observations	18,247	17,672	22,010	50,639	48,070	60,640

This table reports triple-difference estimates of the effects of inclusion in EDGAR on forecast inaccuracy. For the corresponding difference-in-differences models, see the analyst-stock level models in Table 4. We interact treatment with three measures of an analyst’s incentives to engage in strategic behavior, each measured in the quarter before EDGAR inclusion, so not time varying: how focused on retail investors the analyst’s brokerage house is, whether the analyst is ranked as a “star,” and whether the analyst is “affiliated” in the sense of working for a brokerage house that has an investment banking relationship with the firm that is joining EDGAR. (See Table 1A.9 in the Internet Appendix for summary statistics.) *post* equals one in the quarter of joining EDGAR and the next four quarters. Since the triple-interaction variables are measured in the quarter before treatment, they do not vary within firm and their levels effects are absorbed by the firm effects. All specifications are estimated using OLS and include controls (the one-quarter lag of log market cap) and fixed effects for calendar-quarter, fiscal-quarter, and analyst-firm. For variable definitions and details of their construction, see Appendix B. The coefficients are multiplied by 100 for ease of exposition. Heteroscedasticity-consistent standard errors double-clustered at firm and analyst-quarter levels are shown in italics underneath the coefficient estimates. * $p < .1$; ** $p < .05$; *** $p < .01$.

behavior, we expect a larger reduction in optimism and inaccuracy post-EDGAR among affiliated analysts than among unaffiliated analysts. Consistent with this prediction, the triple interaction *Treated × Post × Affiliated* has an economically large negative coefficient in all four specifications, significantly so in two of them.

In sum, the triple-diff results in Table 8 suggest that EDGAR inclusion has a larger effect on the behavior of those analysts the literature regards as most susceptible to strategic considerations. We view these results as consistent with the interpretation that universal access to corporate filings curtails a strategic component of analyst behavior as more investors can more easily verify analyst reports.

3.2 Broker channel

EDGAR gives everyone access to mandatory disclosures, including analysts. What if an analyst (unusually) did not have access to any SEC filings pre-EDGAR? The introduction of EDGAR could then lead her to change

her behavior for nonstrategic reasons, as her information production ability improves. If sufficiently many analysts lack access to SEC filings pre-EDGAR, this could then explain why forecasts become more accurate post-EDGAR (Table 4).

Prior research as well as our own findings cast doubt on the broker channel. Christensen, Heninger, and Stice (2013), who investigate the effect of EDGAR on analysts' information production ability, conclude that "EDGAR has not dramatically transformed the information environment for analysts. They used SEC filing data to inform their earnings forecasts before EDGAR and have continued to do so in the EDGAR era."

As for our own findings, it is not obvious why—if EDGAR did improve analysts' information production ability—their forecasts become less optimistic (Table 4), especially among firms that more aggressively used discretionary accruals pre-EDGAR and whose disclosures are more extensive (Table IA.3) and when they predominantly serve retail clients or the firm is an underwriting client (Table 8, panel A). None of these findings is consistent with the broker channel for the simple reason that improved information production ability has no bearing on optimism, which is explicitly viewed in the literature as reflecting strategic behavior. Nor can the broker channel account for our findings that analyst forecasts become less informative (Table 5) or why analysts drop coverage (Table 3): improved information production ability would suggest that forecasts should become more informative³⁷ and that coverage should increase, all else equal. Finally, the lack of spillovers to the *other* firms an analyst covers is also not obviously consistent with improved information production: if firm *i* joining EDGAR did simply improve the analyst's information set, this should manifest as increased accuracy in the analyst's future-treated stocks (Goldstein and Yang 2017), but that is not what we find (Table IA.7).

Still, we test the broker channel in two ways: using cross-sectional variation in the extent of access to SEC filings pre-EDGAR, and using an institutional quirk of the way EDGAR was rolled out.

3.2.1 Cross-sectional predictions. The broker-channel argument applies more to some analysts (those without access to SEC filings via commercial data feeds) than to others (those with access via commercial vendors). In other words, the argument implies a heterogeneous treatment effect whereby inclusion in EDGAR improves accuracy more among some analysts than among others. Data on which brokers subscribed to data feeds pre-EDGAR

³⁷ A reviewer notes that informativeness could decline even in the broker channel, namely, if EDGAR inclusion leads to either an increase in the number of analysts with correlated information or to an increase in "tipping" to favored clients. We acknowledge these possibilities, though we add that we know of no prior evidence in their support. Our own evidence shows a *reduction* in the number of analysts (Table 3, column 1), especially in cases in which correlated information is arguably most likely (Table 3, column 2). The leveling of the informational playing field that EDGAR was designed to bring about makes tipping less likely in our view, not more likely.

are not publicly available. However, it seems reasonable to assume that there would have been substantial economies of scale in data-feed costs. If so, larger brokers and those covering a larger fraction of the universe of firms could have spread their data-feed costs over a larger quantity of output and so would have been more likely to subscribe to data feeds than smaller ones, all else equal.³⁸

We use two measures of a broker's size (the number of analysts it employs and its annual fee revenue from equity underwriting) and two measures of its breadth of coverage (the fraction of U.S.-listed stocks, by either number or market capitalization, covered by its analysts). All four variables are measured as of the quarter before a firm joins EDGAR. Table 9, panel A, reports the results of triple-difference specifications. We find that the reductions in inaccuracy observed around EDGAR inclusion are *not* concentrated among analysts working for smaller brokers or for brokers covering only a small part of the stock universe, contrary to the broker-channel argument. In fact, the reductions in inaccuracy do not vary significantly with broker size (except, marginally, in column 6, which considers the effect of size as measured by equity underwriting fees on the reduction in long-term inaccuracy).³⁹

As noted, the broker channel cannot explain why, according to Table 4, optimism declines significantly when a firm's filings become publicly available. Table 9, panel B, reports triple-diff specifications for the effect of broker size on the reduction in optimism. Interestingly, we find significantly *larger* reductions in short-term optimism following EDGAR inclusion among analysts at larger brokers and at brokers covering more of the stock universe. This is consistent with strategic behavior to the extent that larger brokers have more reputation capital at stake and so derive a larger benefit from reining in their analysts' strategic behavior when EDGAR inclusion increases the threat of investors detecting strategic behavior, compared to smaller brokers.

3.2.2 Electronic filing versus online access. As noted in Section 1.1, it was never the SEC's intention to put EDGAR filings online; EDGAR was simply intended as an electronic filing system. It took further lobbying and NSF funding to eventually create online access in January 1994. For firms in waves 1–4, therefore, analysts had no reason to expect the public to gain

³⁸ In practice, our interviews with veteran analysts suggest that even smaller brokers considered access to corporate filings mission-critical pre-EDGAR and were willing to pay for it. Moreover, as a last resort, analysts could request filings be faxed over by the company's investor relations team. If so, the broker channel has little empirical merit.

³⁹ While statistically insignificant, each of the eight triple-diff coefficients in panel A is negative. The negative sign directly runs counter to the hypothesis that analyst behavior changed around EDGAR inclusion just because analysts themselves gained access to filings they could not access previously. The only way to square the negative triple-diff estimates with the broker channel is to assume implausibly that while analysts at smaller brokers already had access to SEC filings pre-EDGAR, analysts at larger brokers did not.

Table 9
Triple-difference estimates of the effects of inclusion in EDGAR on forecast inaccuracy

A. Broker channel: Forecast inaccuracy								
	Inaccuracy							
	Short-term				Long-term			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
<i>Treated × Post</i>	−0.414** <i>0.169</i>	−0.405 <i>0.250</i>	−0.434*** <i>0.160</i>	−0.401*** <i>0.154</i>	−0.675** <i>0.270</i>	−0.298 <i>0.368</i>	−0.698*** <i>0.267</i>	−0.752*** <i>0.260</i>
<i>Treated × Post × # analysts</i>	−0.003 <i>0.004</i>				−0.005 <i>0.006</i>			
<i>Treated × Post × Equity fees</i>		−0.009 <i>0.017</i>				−0.040* <i>0.023</i>		
<i>Treated × Post × Coverage(#)</i>			−0.996 <i>1.290</i>				−1.665 <i>2.179</i>	
<i>Treated × Post × Coverage(\$)</i>				−0.437 <i>0.421</i>				−0.306 <i>0.688</i>
Controls?	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Calendar quarter FE?	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Fiscal quarter FE?	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Analyst-firm FE?	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
<i>R-squared (%)</i>	61.6	59.5	61.6	61.6	71.3	69.1	71.3	71.3
No. of firms	2,244	2,062	2,244	2,244	2,826	2,606	2,826	2,826
No. of observations	21,723	18,247	21,723	21,723	59,920	50,639	59,920	59,920

For the corresponding difference-in-differences models, see the analyst-stock level models in Table 4. We interact treatment with four measures of brokerage house size, each measured in the quarter before EDGAR inclusion, so not time varying: the number of analysts the broker employs (*# analysts*), the log of the broker's annual fee income from underwriting equity issues (*Equity fees*), and the fraction of the universe of U.S.-listed stocks the broker's analysts cover, by number (*Coverage(#)*) and by market cap (*Coverage(\$)*). (See Table IA.9 in the Internet Appendix for summary statistics.) *Post* equals one in the quarter of joining EDGAR and the next four quarters. Since the triple-interaction variables are measured in the quarter before treatment, they do not vary within firm and their levels effects are absorbed by the firm effects. All specifications are estimated using OLS and include controls (the one-quarter lag of log market cap) and fixed effects for calendar-quarter, fiscal-quarter, and analyst-firm. For variable definitions and details of their construction, see Appendix B. The coefficients are multiplied by 100 for ease of exposition. Heteroscedasticity-consistent standard errors double-clustered at firm and analyst-quarter levels are shown in italics underneath the coefficient estimates. **p* < .1; ***p* < .05; ****p* < .01.

(Continued)

access to electronic filings. At the same time, some analysts may have benefited from EDGAR filings in the form of reduced information acquisition costs. This institutional quirk allows us to investigate the strategic-analyst and broker channels as follows. In the strategic-analyst channel, we expect an analyst to change her behavior not when wave 1–4 firms join EDGAR (as she had no reason to expect an increase in the risk of detection) but when their filings are unexpectedly put online in Jan. 1994. In the broker channel, we expect either no response on either date (if the analyst already had access to SEC filings) or a response when wave 1–4 firms join EDGAR (if the analyst lacked access to SEC filings before), but not when their filings subsequently go online. The only group of analysts who would plausibly improve their forecasts in January 1994 for nonstrategic reasons are those who had to wait for the NSF-funded online access (i.e., presumably those at smaller brokers).

Table 9, panel C, restricts the triple-diff specification used in Table 9 to waves 1–4, using the number of analysts to capture broker size. We report two models, a placebo model (which treats the quarter a firm joins EDGAR as an electronic filer as the treatment event) and our baseline model (which

Table 9
Triple-difference estimates of the effects of inclusion in EDGAR on forecast inaccuracy (continued)

	B. Forecast optimism and broker size							
	Optimism							
	Short-term				Long-term			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
<i>Treated × Post</i>	−0.384**	−0.557**	−0.409**	−0.383**	−0.891***	−0.998**	−0.879***	−0.927***
<i>Treated × Post × # analysts</i>	0.184	0.279	0.174	0.166	0.293	0.398	0.287	0.279
<i>Treated × Post × Equity fees</i>	−0.008**				−0.009			
	0.004				0.007			
<i>Treated × Post × Coverage(#)</i>		−0.009				−0.013		
		0.019				0.025		
<i>Treated × Post × Coverage(\$)</i>			−3.184**				−3.725	
			1.467				2.546	
<i>Treated × Post × Coverage(\$)</i>				−1.168**				−1.052
				0.485				0.835
Controls?	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Calendar quarter FE?	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Fiscal quarter FE?	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Analyst-firm FE?	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
R-squared (%)	56.8	54.2	56.8	56.8	67.5	65.1	67.5	67.5
No. of firms	2,244	2,062	2,244	2,244	2,826	2,606	2,826	2,826
No. of observations	21,723	18,247	21,723	21,723	59,920	50,639	59,920	59,920

This table reports triple-difference estimates of the effects of inclusion in EDGAR on forecast optimism. For the corresponding difference-in-differences models, see the analyst-stock level models in Table 4. We interact treatment with four measures of brokerage house size, each measured in the quarter before EDGAR inclusion, so not time varying: the number of analysts the broker employs (*# analysts*), the logarithm of the broker's annual fee income from underwriting equity issues (*Equity fees*), and the fraction of the universe of U.S.-listed stocks the broker's analysts cover, by number (*Coverage(#)*) and by market cap (*Coverage(\$)*). (See Table IA.9 in the Internet Appendix for summary statistics.) *post* equals one in the quarter of joining EDGAR and the next four quarters. Since the triple-interaction variables are measured in the quarter before treatment, they do not vary within firm and their levels effects are absorbed by the firm effects. All specifications are estimated using OLS and include controls (the one-quarter lag of log market cap) and fixed effects for calendar-quarter, fiscal-quarter, and analyst-firm. For variable definitions and details of their construction, see Appendix B. The coefficients are multiplied by 100 for ease of exposition. Heteroscedasticity-consistent standard errors double-clustered at firm and analyst-quarter levels are shown in italics underneath the coefficient estimates. * $p < .1$; ** $p < .05$; *** $p < .01$.

(Continued)

treats online access in 1994Q1 as the treatment event). The placebo model in columns 1–4 shows that neither optimism nor accuracy changes significantly when firms join EDGAR; optimism and inaccuracy only decline significantly from January 1994, when filings unexpectedly go online, the more so the more analysts work at the brokerage firm.⁴⁰ The baseline model in columns 5–8 confirms these patterns. Optimism and inaccuracy decline significantly when filings go online, with the size of these improvements increasing in brokerage size. Before January 1994, there is no evidence of changes in forecast behavior.

In sum, we find no change in behavior when firms in waves 1–4 join EDGAR. Instead, analysts (and especially those working at larger brokers) only change their behavior after filings are put online, that is, when *everyone* gains access to these filings. This pattern is consistent with the strategic-analyst channel.

⁴⁰ For firms joining EDGAR in wave 4 on December 6, 1993 (and maybe those joining EDGAR in wave 3 on Oct. 4, 1993), the placebo test may lack power to disentangle the effects of joining EDGAR and of filings becoming available online on January 17, 1994. Our results are robust to restricting the placebo test to firms joining in waves 1 and 2; see Table IA.10 in the Internet Appendix.

Table 9
Triple-difference estimates of the effects of inclusion in EDGAR on forecast inaccuracy (continued)

	Placebo sample				Baseline sample			
	Optimism		Inaccuracy		Optimism		Inaccuracy	
	Short-term	Long-term	Short-term	Long-term	Short-term	Long-term	Short-term	Long-term
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
<i>Treated</i> × Pre-Jan. 17, 1994					−0.318	−0.557	−0.338	−0.477
					<i>0.515</i>	<i>0.318</i>	<i>0.437</i>	<i>0.290</i>
×# analysts					−0.009	0.006	−0.010	0.005
					<i>0.008</i>	<i>0.005</i>	<i>0.007</i>	<i>0.004</i>
<i>Treated</i> × Post	−0.108	−0.537	−0.180	−0.545	−0.329	−1.362***	−0.503**	−1.139***
	<i>0.346</i>	<i>0.479</i>	<i>0.315</i>	<i>0.470</i>	<i>0.240</i>	<i>0.357</i>	<i>0.230</i>	<i>0.305</i>
×# analysts	−0.005	0.002	−0.004	0.002	−0.018**	−0.022**	−0.014**	−0.014
	<i>0.006</i>	<i>0.004</i>	<i>0.005</i>	<i>0.004</i>	<i>0.007</i>	<i>0.010</i>	<i>0.006</i>	<i>0.008</i>
<i>Treated</i> × Post-Jan. 17, 1994	−0.440**	−1.712***	−0.388**	−1.179***				
	<i>0.193</i>	<i>0.344</i>	<i>0.171</i>	<i>0.318</i>				
×# analysts	−0.012**	−0.018***	−0.009*	−0.013**				
	<i>0.006</i>	<i>0.006</i>	<i>0.005</i>	<i>0.005</i>				
R-squared (%)	52.2	62.0	58.1	66.3	56.0	69.0	63.0	74.5
No. of firms	1,303	1,500	1,303	1,500	1,161	1,401	1,161	1,401
No. of observations	17,214	55,098	17,214	55,098	12,021	37,130	12,021	37,130

The table reports triple-difference estimates of the effects of joining EDGAR as an electronic filer or of online access to EDGAR filings on forecast optimism and inaccuracy. Treated firms are restricted to those joining EDGAR in waves 1–4; control firms are nearest-neighbor propensity-score matched on equity market capitalization (in levels and logs), number of analysts (in logs and lags), and fiscal quarter using a 0.05 caliper. In the “placebo sample” in columns 1–4, we include data from a nine-fiscal quarter window centered on the fiscal quarter in which a treated firm joins EDGAR as an electronic filer. In the “baseline sample” in columns 5–8, we include data from a nine-fiscal quarter window centered on the fiscal quarter in which a treated firm’s EDGAR filings are first made available online (Jan. 17, 1994). The placebo sample is larger than the baseline sample because shifting “event quarter 0” earlier (as the placebo sample does, by pretending that treatment occurs when a firm joins EDGAR rather than when, as in our baseline sample, its filings become publicly available) opens up a larger set of valid future-treated controls. The placebo sample excludes a small number of wave 4 firms that joined EDGAR in Dec. 1993 and whose filings went online in the same fiscal quarter; for these firms, the baseline treatment and the online-access treatment occur in the same fiscal quarter and so cannot be disentangled. *post* equals one in the respective placebo or treatment quarter and the next four quarters. We include interactions with the number of analysts the broker employs (# analysts) to proxy for brokerage house size, measured as of the quarter before EDGAR inclusion. (See Table IA.9 in the Internet Appendix for summary statistics.) Since these triple-interaction variables are measured in the quarter before treatment, they do not vary within firm and their levels effects are absorbed by the firm effects. All specifications are estimated using OLS and include controls (the one-quarter lag of log market cap) and fixed effects for calendar-quarter, fiscal-quarter, and analyst-firm pair. For variable definitions and details of their construction, see Appendix B. The coefficients are multiplied by 100 for ease of exposition. Heteroscedasticity-consistent standard errors are shown in italics underneath the coefficient estimates. They are double-clustered at the firm and analyst-quarter level. * $p < .1$; ** $p < .05$; *** $p < .01$.

4. Conclusions

A rich literature documents that sellside analysts engage in strategic behavior rather than providing objective information to buy-side clients: analysts are prone to biasing earnings forecasts, to inflating recommendations, and to suspending coverage rather than issuing unflattering reports when a firm is doing poorly. We provide evidence that permits the interpretation that analysts’ strategic behavior is constrained by investors’ ability to verify analyst reports. Using a randomly assigned shock, we find that free, timely, and equal access to firms’ mandatory disclosures results in analysts making earnings forecasts that are less optimistic and more accurate. The shock thins the ranks of analysts covering a given firm as analysts whose reports add little value when

corporate filings become freely available exit, consistent with the model of Dugast and Foucault (2018). It also results in analyst forecasts moving share prices by less as investors gain access to better conditioning information. At the same time, analysts inflate their stock recommendations by more. Overall, free, timely, and equal access to corporate filings improves market quality, as measured by liquidity, volatility, and speed with which earnings news is incorporated in stock prices.

The natural experiment we use is the SEC's rollout of the EDGAR system in the early 1990s. We take seriously the possibility that EDGAR could have changed analyst behavior because it improved analysts' own access to SEC filings but find no support for it in the historical record, in prior literature, or in our own empirical tests. We also investigate the possibility that behavior may have changed because firms responded to being included in EDGAR by changing reporting practices but again find no support for it. Instead, based on the findings that analyst behavior changed only when the risk of detection became real and that it changed the most among analysts who had the greatest incentive to change their strategic behavior, we favor the interpretation that analysts changed a strategic component of their forecasts.

We view the information-economic effects of EDGAR inclusion as a reduction in investors' costs of verifying the veracity of information provided by analysts *ex post*. Our results suggest that reduced verification costs constrain analysts' ability to strategically skew their forecasts in ways that benefit themselves or their brokerage-firm employers. These findings highlight the importance of verification costs in the game analysts and investors play. The nature of our experiment is such that it can plausibly be interpreted to vary verification costs for a subset of investors (i.e., retail and small institutional investors), from arguably something approaching infinity to something much closer to zero. Seen through this lens, we interpret the observed changes in analyst behavior as indicating that free, timely, and equal access to corporate information improves investors' ability to verify analyst reports *ex post*, which in turn constrains analysts' strategic behavior *ex ante*. Our interpretation fits well with theory models that view reputational concerns as helping to discipline analysts and encouraging truthful communication (Benabou and Laroque 1992; Meng 2015).

Our finding that analysts issue more bullish recommendations after a firm joins EDGAR is consistent with the interpretation that analysts sought to continue to be "nice" to firms post-EDGAR, perhaps in the hope of privileged access to senior executives (a practice Reg FD, introduced in Oct. 2000, seeks to curtail). Unlike earnings forecasts, recommendations lack a specific time horizon and an unambiguous *ex post* benchmark against which their accuracy can be judged and so are considered less verifiable *ex post* than forecasts. As a result, as Lin and McNichols (1998) note, manipulation of "a recommendation is more difficult for investors to detect than manipulation of an earnings forecast." The introduction of EDGAR may thus have made little difference to

investors' ability to police analysts' recommendation behavior.⁴¹ It is possible, therefore, that EDGAR induced analysts to shift their management-pleasing behavior from friendly forecasts to friendly recommendations. Notably, we find no change in recommendations among Value Line analysts, whose behavior the literature considers less prone to strategic incentives.

Partly for this reason, we are not claiming that ex post verification by investors is sufficient to eliminate ex ante strategic behavior among analysts entirely. Indeed, subsequent interventions by the SEC, including Regulation FD in 2000 and the Global Settlement in 2003, suggest that concerns about strategic behavior have persisted post-EDGAR, especially as regards analyst recommendations.

Our findings speak to the interplay between disclosure and information intermediaries, such as analysts, in shaping firms' external information environments. We find that greater access to SEC filings and analyst coverage are substitutes in our setting, consistent with theoretical models showing that greater disclosure could discourage private information production (Gao and Liang 2013; Banerjee, Davis, and Gondhi 2018) and with the finding in Goldstein, Yang, and Zuo (2020) and Bird et al. (2021) that EDGAR reduced the investment- Q sensitivity, which the authors conjecture reflects a crowding out of external information production. We leave to future research whether the partial crowding out of information production by analysts we document improves investor welfare on net

⁴¹ Indeed, Ljungqvist et al. (2007) show that it is primarily institutional investors who moderate analysts' tendency to issue inflated recommendations, not the retail investors who arguably benefited the most from EDGAR.

Appendix A

Table A.1
EDGAR phase-in waves

Phase-in wave no.	SEC designation	Preliminary phase-in date (SEC Release 33-6977)	Revised phase-in date (SEC Release 33-7122)	Online date	All listed U.S. firms		Treated firms	
					No. of firms	Mean market cap (\$m)	No. of firms	Mean market cap (\$m)
1	CF-01	April 26, 1993	April 26, 1993	January 17, 1994	105	8,428.5	23	276.4
2	CF-02	July 19, 1993	July 19, 1993	January 17, 1994	405	4,586.8	45	576.6
3	CF-03	October 4, 1993	October 4, 1993	January 17, 1994	418	964.6	182	327.7
4	CF-04	December 6, 1993	December 6, 1993	January 17, 1994	596	338.5	443	161.7
5	CF-05	August 1994	January 30, 1995	January 30, 1995	664	198.6	491	217.1
6	CF-06	November 1994	March 6, 1995	March 6, 1995	566	91.4	460	87.4
7	CF-07	May 1995	May 1, 1995	May 1, 1995	458	97.1	372	96.1
8	CF-08	August 1995	August 7, 1995	August 7, 1995	246	79.1		
9	CF-09	November 1995	November 6, 1995	November 6, 1995	132	191.1		
10	CF-10	May 1996	May 6, 1996	May 6, 1996	905	356.9		
All					4,495	879.6	2,016	171.7

The table provides a breakdown of the universe of listed U.S. firms and of the sample of treated firms by EDGAR phase-in wave. Listed U.S. firms are those listed on the NYSE, NASDAQ, or AMEX with CRSP share codes of 10 or 11. Treated firms require the existence of a valid control firm using a nearest-neighbor propensity-score method matching on equity market capitalization (in levels and logs), analyst coverages (in logs and lags), and fiscal quarter. Only matches in the common support are considered valid, using a 0.05 caliper. Market cap is measured in the fiscal quarter prior to inclusion in EDGAR.

Appendix B Variable Definitions

Stock-level measures

analysts is the number of analysts who issue earnings-per-share forecasts for a firm in a fiscal quarter, counting unique I/B/E/S analyst identifiers (I/B/E/S unadjusted detail file variable *analysts*).

days is the average number of days between analysts' earnings forecasts for a firm in a fiscal quarter and the earnings announcement date. We compute, for each analyst making an earnings forecast in a fiscal quarter, the number of days between the forecast date and the earnings announcement date. We then average the number of days across analysts following a firm. The variable is separately computed for short-term and long-term forecasts. Short-term forecasts are those made for the next fiscal quarter ($fpi=7$); long-term forecasts are those made for the current fiscal year ($fpi=1$).

Abnormal volume is the quarterly volume ratio constructed following Barber and Odean (2008). It is defined as $(V_{i,t}/V_i)/(V_{m,t}/V_m)$, where $V_{i,t}$ is average daily trading volume for firm i in fiscal quarter t , V_i is average daily trading volume for firm i in fiscal quarter $t-1$, $V_{m,t}$ is average daily market trading volume in quarter t , and V_m is average daily market trading volume in quarter $t-1$. Trading volume is defined as the number of shares traded (CRSP variable *vol*) multiplied by the daily closing price (CRSP variable *prc*). Trading volume on Nasdaq is adjusted using the Gao and Ritter (2010) procedure. Market trading volume is calculated using all CRSP common stocks (share code 10 or 11).

Abnormal volume (retail) is the quarterly turnover by the retail customers of a large discount brokerage firm using the data of Barber and Odean (2000). It is defined as the total number of buy and sell trades divided by the number of shares outstanding at the previous quarter-end (CRSP variable *shrout*).

AIM is the natural logarithm of one plus Amihud's (2002) illiquidity measure. We use daily CRSP data to calculate the ratio of absolute return to dollar volume, $[1,000,000 \times |ret|/(|prc| \times vol)]$, for each trading day in a fiscal quarter. We then average over the quarter and take logs. Trading volume on Nasdaq is adjusted using the Gao and Ritter (2010) procedure.

Breaks in streaks of earnings increases is an indicator variable set equal to one if firm i 's quarter t earnings (Compustat variable niq) decrease after having increased in each of the previous four quarters. This definition follows that of Andreou, Louca, and Petrou (2017).

DA (Jones) is firm i 's discretionary accruals in fiscal quarter t obtained from a modified Jones model following Dechow, Sloan, and Sweeney (1995). The modified Jones model is specified as $TA_{it}/ASSET_{it-1} = \beta_0 + \beta_1 1/ASSET_{it-1} + \beta_2 \Delta REV_{it}/ASSET_{it-1} + \beta_3 PPE_{it}/ASSET_{it-1} + \varepsilon_{it}$, where TA_{it} is total accruals, defined as earnings before extraordinary items and discontinued operations (Compustat variable ibq) minus operating cash flows (Compustat variable $oancfy$), $ASSET_{it-1}$ is lagged total assets (Compustat variable atq), ΔREV_{it} is the change in quarterly revenue (Compustat variable $saleq$), and PPE_{it} is gross property, plant, and equipment (Compustat variable $ppeqtq$). Jones discretionary accruals is defined as $DA_{it} = (TA_{it}/ASSET_{it-1}) - NA_{it}$, where $NA_{it} = \hat{\beta}_0 + \hat{\beta}_1 1/ASSET_{it-1} + \hat{\beta}_2 (\Delta REV_{it} - \Delta AR_{it})/ASSET_{it-1} + \hat{\beta}_3 PPE_{it}/ASSET_{it-1}$ and AR_{it} is accounts receivable (Compustat variable $rectq$).

DA (Kothari) is the performance-matched discretionary accruals in a fiscal quarter following Kothari, Leone, and Wasley (2005), defined as a firm's discretionary accruals from a modified Jones model minus that of a matched firm in the same Fama-French 48 industry with the closest return on assets.

Dispersion is the standard deviation of analysts' earnings forecasts made in fiscal quarter t (I/B/E/S variable $stdev$), scaled by the end-of-quarter stock price (CRSP variable prc). I/B/E/S data are obtained from the unadjusted summary history files. Short-term dispersion is based on forecasts made for fiscal quarter $t+1$ ($fpi=7$); long-term dispersion is based on forecasts made for the current fiscal year ($fpi=1$). See Lehavy, Li, and Merkley (2011) for further details.

DQ is a firm's quarterly "disclosure quality" score. It captures the level of disaggregation in its financial reporting by counting the number of nonmissing Compustat line items. The score ranges from 0 to 1, where 0 (1) equals the lowest (highest) disclosure quality. It is computed separately for the income statement and the balance sheet and averaged to the firm level. See Chen, Miao, and Shevlin (2015) for further details.

Earnings restatement is an indicator variable set equal to one if the absolute difference between firm i 's quarter t I/B/E/S earnings per share (variable $value$) and Compustat earnings per share (variable $epspxq$) is equal to or greater than 0.015. This definition follows that of Livnat and Mendenhall (2006).

Effective tick is the quarterly average of Goyenko, Holden, and Trzcinka's (2009) effective tick measure. Using daily CRSP data (CRSP variables prc and vol) and based on end-of-day price clustering, we calculate an average effective spread over the quarter as the probability-weighted average of each effective spread size deflated by the stock price.

Fraction zero-return is the fraction of trading days with zero or missing returns in a fiscal quarter. See Lesmond, Ogden, and Trzcinka (1999) and Goyenko, Holden, and Trzcinka (2009) for further details.

Idiosyncratic volatility is the standard deviation of regression residuals from a Fama-French three-factor model using daily stock returns (CRSP variable ret) in a firm-fiscal quarter, measured following Ang et al. (2006).

Inaccuracy is the average absolute difference between analysts' earnings forecasts and realized earnings for a firm in a fiscal quarter. Following Hong and Kubik (2003), we compute, for each analyst making an earnings forecast in a fiscal quarter, the absolute difference between realized earnings and the forecast, scaled by the previous fiscal quarter-end share price (CRSP monthly file variable prc). We are careful to compare diluted forecasts to diluted earnings (Compustat variables $epsfxq$ and $epsfx$ for quarterly and annual earnings, respectively) and primary forecasts

to primary earnings (Compustat variables *epspxq* and *epspx* for quarterly and annual earnings, respectively). We then average the absolute differences across analysts following a firm. Short-term forecast inaccuracy is based on forecasts made for the next fiscal quarter ($fpi = 7$); long-term inaccuracy is based on forecasts made for the current fiscal year ($fpi = 1$).

Informativeness is the fraction of cumulative daily absolute abnormal returns that can be attributed to analyst forecasts in a fiscal quarter. Following Lehavy, Li, and Merkley (2011), the measure is defined as $\sum_{d=1}^{NREVS} |R_{i,d} - Dec\ ret_{i,d}| / \sum_{d=1}^D |R_{i,d} - Dec\ ret_{i,d}| \cdot NREVS$ is the number of trading days for which there is at least one analyst forecast in the I/B/E/S detail history file. D is the number of trading days in a quarter. $R_{i,d}$ is the daily return of firm i on day d (CRSP variable *ret*). $Dec\ ret_{i,d}$ is the CRSP size-decile portfolio return (variable *decret*).

Meet-or-beat is an indicator variable set equal to one if a firm's *EPS* is both greater than and within 1 cent of the median analyst's earnings forecast.

Optimism is the average difference between analysts' earnings forecasts and realized earnings for a firm in a fiscal quarter. Following Abarbanell and Lehavy (2003), we compute, for each analyst making an earnings forecast in a fiscal quarter, the difference between realized earnings and the forecast, scaled by the previous fiscal quarter-end share price (CRSP monthly file variable *prc*). We are careful to compare diluted forecasts to diluted earnings (Compustat variables *epsfxq* and *epsfx* for quarterly and annual earnings, respectively) and primary forecasts to primary earnings (Compustat variables *epspxq* and *epspx* for quarterly and annual earnings, respectively). We then average the differences across analysts following a firm. Short-term forecast optimism is based on forecasts made for the next fiscal quarter ($fpi = 7$); long-term optimism is based on forecasts made for the current fiscal year ($fpi = 1$).

Price impact of recommendation changes is the average stock return around analyst recommendation revisions for a firm in a fiscal quarter. Following Mikhail, Walther, and Willis (2004), we compute the 3-day cumulative Fama-French three-factor (FF3) adjusted return around each recommendation change by taking a long (short) position in upgrades (downgrades) and then average within firm-fiscal quarter. An upgrade (downgrade) is defined as a decrease (increase) in an analyst's I/B/E/S recommendation level (variable *ireccd*). The daily FF3-adjusted return is measured as the difference in the daily stock return (CRSP variable *ret*) and the daily stock return predicted by FF3. We estimate the FF3 factor loadings with a time-series regression with one quarter of daily returns ending on the last trading day of the firm's previous fiscal quarter. Recommendations data come from Ljungqvist, Marston, and Wilhelm (2009), who note that "the overlap between I/B/E/S and First Call is only 46.8% over [their] sample period" and therefore combine the two databases. Their data also correct for the widespread and apparently nonrandom problems in the I/B/E/S historical recommendations database identified by Ljungqvist, Malloy, and Marston (2009).

Price convergence is the absolute cumulative abnormal return around a firm's quarterly earnings announcement. Following Heflin, Subramanyam, and Zhang (2003), we compute $\left| \prod_{d=-30}^2 (1 + AR_{i,q,d}) - 1 \right|$ for each firm i and fiscal quarter t , from 30 days before the earnings announcement date to 2 days after. Daily abnormal returns are CAPM-adjusted. Earnings announcement dates are from Compustat (variable *rdq*).

Recommendation strength is the average of outstanding analyst recommendations for a firm in a fiscal quarter. Recommendations are reverse-scored using I/B/E/S variable *ireccd* such that values 1, 2, 3, 4, and 5 correspond to a sell, underperform, hold, buy, and strong buy recommendation, respectively.

Reporting richness is estimated following Coles, Daniel, and Naveen (2008). Specifically, for each firm/fiscal-year observation, we compute a factor score based on the number of business segments (from the Compustat segments database), the logarithm of annual sales (Compustat item

sale), and book leverage (Compustat item *dltt* divided by Compustat item *at*). The factor score for a firm/fiscal-year observation is a linear combination of the transformed-to-standard-normal values of these three variables.

Revision response coefficient is the estimated stock price response to analyst forecast revisions for a firm in a fiscal quarter. Following Park and Stice (2000), we first compute analyst forecast revisions by analysts following the firm in a quarter, defined as the difference in consecutive earnings forecasts by each analyst (I/B/E/S variable *value*), scaled by the stock price 2 days before the revision (CRSP variable *prc*). Next, we regress the 3-day cumulative FF3-adjusted returns around forecast revisions on analyst forecast revisions. Daily FF3-adjusted return is defined as the difference in the daily stock return (CRSP variable *ret*) and the return from the FF3 model. The revision response coefficient is the estimated slope coefficient from this regression.

ROA is the firm's diluted quarterly earnings per share (Compustat variable *epsfxq*), scaled by the previous quarter-end stock price (CRSP variable *prc*).

Volatility at earnings announcement is the annualized daily DGTW-adjusted return volatility in a 3-day window centered on a firm's earnings announcement date, following Kelly and Ljungqvist (2012). Earnings announcement dates are from Compustat (variable *rdq*). Daily returns are from CRSP (variable *ret*).

Volume at earnings announcement is the sum of the CRSP daily logarithm of trading volume (variable *vol*) in a 3-day window centered on a firm's earnings announcement date. Earnings announcement dates are from Compustat (variable *rdq*). CRSP trading volume is adjusted using the Gao and Ritter (2010) algorithm.

Analyst-stock level measures

days is the average number of days between an analyst's earnings forecasts for a firm and the earnings announcement date. We compute the number of days between the forecast date of each forecast the analyst makes for that firm that quarter and the earnings announcement date. We then average the number of days for that analyst for that firm in that quarter. The variable is separately computed for short- and long-term forecasts. Short-term forecasts are those made for the next fiscal quarter (*fpi* = 7); long-term forecasts are those made for the current fiscal year (*fpi* = 1).

Affiliated analyst is an indicator variable set equal to one if the analyst works for a brokerage house that underwrote any of the firm's equity or debt issues in the previous 3 years. In our triple-diff specifications, affiliation is measured in the fiscal quarter before the focal firm joins EDGAR.

Boldness is the absolute difference between an analyst's most recent earnings forecast for a firm in the first 2 months of a quarter and the average consensus earnings forecast made by all other analysts covering the firm. Short-term forecast boldness is based on forecasts made for the next fiscal quarter (*fpi* = 7); long-term boldness is based on forecasts made for the current fiscal year (*fpi* = 1).

Coverage is an indicator variable set equal to one if an analyst issues an earnings forecast for firm *i* in fiscal quarter *t*, according to the I/B/E/S unadjusted detail history file.

Inaccuracy is the average absolute difference between an analyst's earnings forecasts and realized earnings for a firm in a fiscal quarter. Following Hong and Kubik (2003), we compute the absolute difference between realized earnings and each forecast the analyst makes for that firm that quarter, scaled by the firm's previous fiscal quarter-end share price (CRSP monthly file variable *prc*). We are careful to compare diluted forecasts to diluted earnings (Compustat variables *epsfxq* and *epsfx* for quarterly and annual earnings, respectively) and primary forecasts to primary earnings (Compustat variables *epspxq* and *epspx* for quarterly and annual earnings, respectively). We then average the absolute scaled differences for that analyst for that firm in that quarter. Short-term

forecast inaccuracy is based on forecasts made for the next fiscal quarter ($fpi=7$); long-term inaccuracy is based on forecasts made for the current fiscal year ($fpi=1$).

Optimism is the average difference between an analyst's earnings forecasts and realized earnings for a firm in a fiscal quarter. Following Abarbanell and Lehavy (2003), we compute the difference between realized earnings and each forecast the analyst makes for that firm that quarter, scaled by the firm's previous fiscal quarter-end share price (CRSP monthly file variable *prc*). We are careful to compare diluted forecasts to diluted earnings (Compustat variables *epsfxq* and *epsfx* for quarterly and annual earnings, respectively) and primary forecasts to primary earnings (Compustat variables *epspxq* and *epspx* for quarterly and annual earnings, respectively). We then average the scaled differences for that analyst for that firm in that quarter. Short-term forecast optimism is based on forecasts made for the next fiscal quarter ($fpi=7$); long-term optimism is based on forecasts made for the current fiscal year ($fpi=1$).

Recommendation strength is an analyst's outstanding recommendation for a firm in a fiscal quarter. Recommendations are reverse-scored using I/B/E/S variable *ireccd* such that values 1, 2, 3, 4, and 5 correspond to a sell, underperform, hold, buy, and strong buy recommendation, respectively.

Analyst-level measures

Star analyst is an indicator variable set equal to one if the analyst is ranked an all-star analyst by either the *Wall Street Journal* (in the *Journal's* June rankings immediately preceding the fiscal quarter before the focal firm joins EDGAR) or *Institutional Investor* magazine (in the October rankings immediately preceding the fiscal quarter before the focal firm joins EDGAR).

Broker-level measures

#analysts is the number of analysts working at each broker according to I/B/E/S, measured in the fiscal quarter before the focal firm joins EDGAR.

Coverage(\$) is the total market capitalization of all firms covered by a broker's analysts divided by the total market capitalization of all firms in I/B/E/S, both measured in the fiscal quarter before the focal firm joins EDGAR.

Coverage(#) is the number of all firms covered by a broker's analysts divided by the total number of firms in I/B/E/S, measured in the fiscal quarter before the focal firm joins EDGAR.

Equity fees is the natural logarithm of a broker's annual revenue from equity underwriting (across all its clients), measured in the fiscal quarter before the focal firm joins EDGAR.

Retail focus is the ratio (expressed as a percentage) of the number of retail representatives to the total number of registered representatives at each broker, measured in the fiscal quarter before the focal firm joins EDGAR. The data come from the Securities Industry Association's yearbooks.

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